
NAND Flash memory

Samsung Electronics, co., Ltd
Flash design team

2010. 05. 07

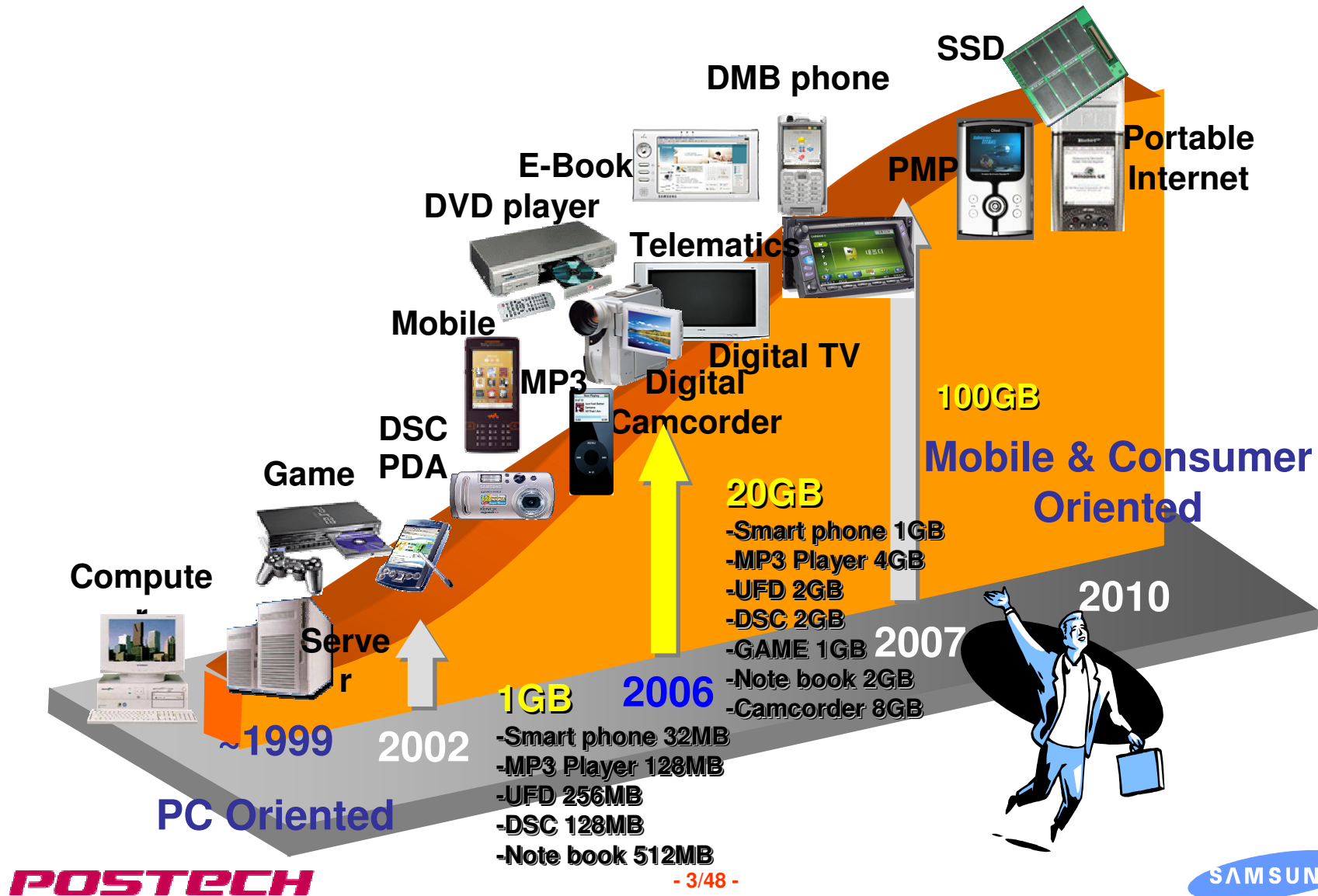
Kihwan Choi

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- **Introduction**
- **Flash memory 101**
- **Basic operations**
- **Current issues & approach**
- **In the near future**



NAND Flash Application



"Flash memory" in Wikipedia

- Flash memory is a **non-volatile computer storage** technology that can be **electrically erased and reprogrammed**.
- Flash memory offers **fast read access times** (although not as fast as volatile DRAM memory used for main memory in PCs) and **better kinetic shock resistance** than hard disks.
- Flash memory (both NOR and NAND types) was invented by Dr. Fujio Masuoka while working for Toshiba circa 1980. According to Toshiba, **the name "flash" was suggested** by Dr. Masuoka's colleague, Mr. Shoji Ariizumi, **because the erasure process of the memory contents reminded him of the flash of a camera**.
- NAND flash also uses floating-gate transistors, but they are connected in a way that **resembles a NAND gate** : several transistors are connected in series, and only if all word lines are pulled high (above the transistors' V_T) is the bit line pulled low.



"Flash memory" in Wikipedia

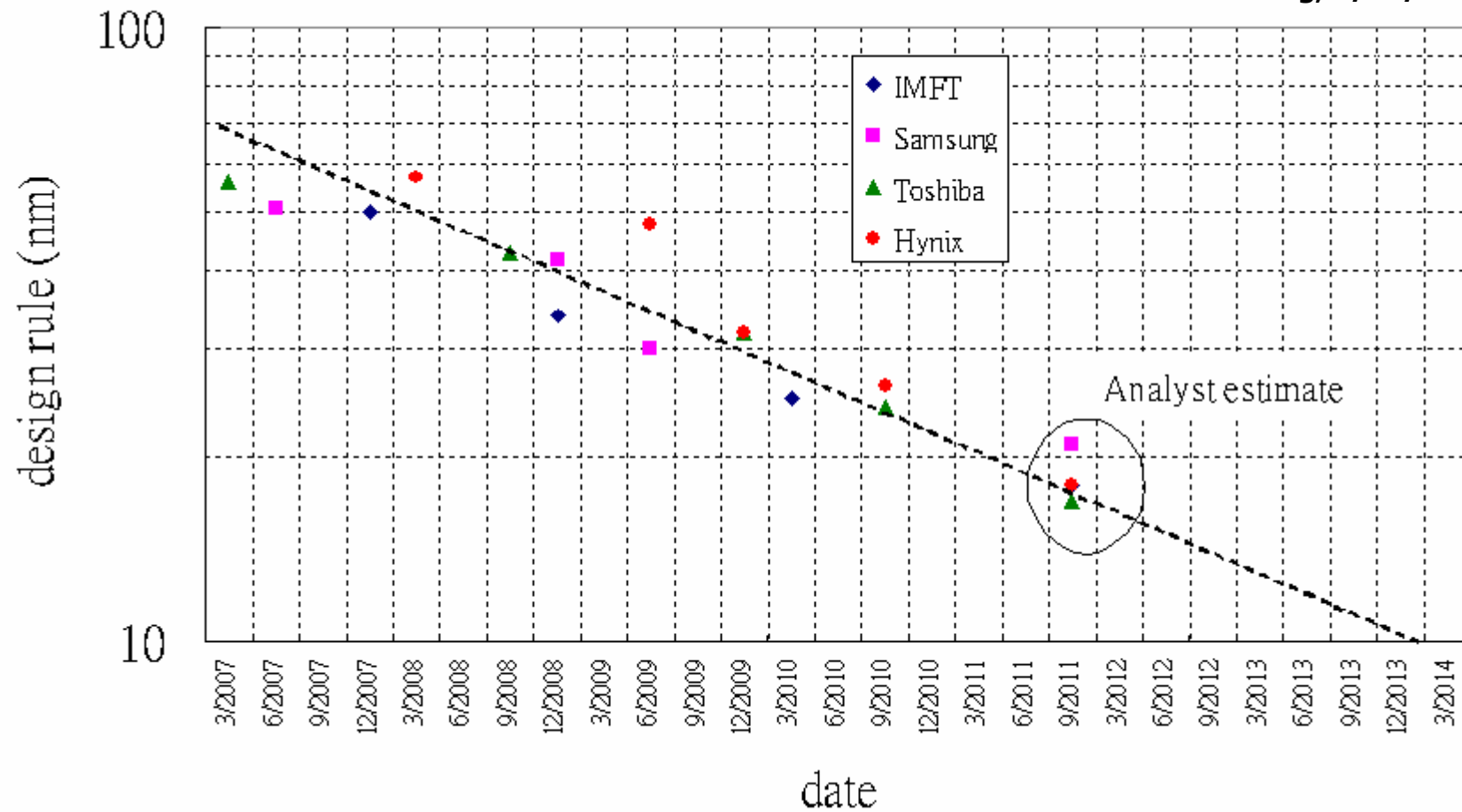
- To read, most of the word lines are pulled up above the VT of a programmed bit, while one of them is pulled up to just over the VT of an erased bit. The series group will conduct (and pull the bit line low) if the selected bit has not been programmed. **NAND flash uses tunnel injection for writing and tunnel release for erasing.**
- One limitation of flash memory is that although it can be read or programmed a byte or a word at a time in a random access fashion, it must be erased a "block" at a time.
- Another limitation is that flash memory has a finite number of erase-write cycles. Most commercially available flash products are guaranteed to withstand around 100,000 write-erase-cycles, before the wear begins to deteriorate the integrity of the storage



"Flash memory" in Wikipedia

- Technology scaling down

Reference: EETimes article on NAND scaling, 3/22/2010



Major players

Table 4. NAND Market Share

2009	Sales (\$M)	Share (%)	GB Equivalent (M)	Share (%)
Samsung	5,337	37.10	2,007	31.50
Toshiba	3,570	24.80	1,689	26.50
SanDisk	1,997	13.90	1,181	18.50
Micron	1,203	8.40	664	10.40
Hynix	1,066	7.40	347	5.40
Intel	826	5.70	465	7.30
Numonyx	330	2.30	18	0.30
Spansion	37	0.30	3	0.00
Powerchip	3	0.00	2	0.00
Renesas	2	0.00	-	0.00
Total	14,371	100	6,375	100

Source: Gartner (March 2010)

Now in the market ...



Intel, Micron and IMFT announce world's first 25-nm NAND technology

February 2, 2010, By Sanjeev Ramachandran in Hardware

The NAND flash production scene has received a shot in the arm with Intel Corporation and Micron Technology making it public that the world's first 25-nanometer (nm) NAND technology is now on stream. Significantly enough, the 25nm process is the smallest NAND technology as well as the smallest semiconductor technology in the world

Hynix Develops 26nm NAND Flash Memory



Tuesday, February 09, 2010

South Korea's Hynix Semiconductor Inc., the world's second-largest memory chipmaker, said Tuesday that it has developed a 26-nanometer based NAND flash memory chip. The company is the world's second flash memory maker to apply the below 30-nanometer technology. Mass production of the new memory will start in in July.

Toshiba readies sub-25nm flash memory chip production

By Jose Vilches, TechSpot.com Published: April 5, 2010, 12:14 PM EST

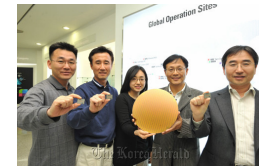


The company produces 32nm and 43nm memory chips, but the plan is to begin production on "sub-25nm" chips that would enable larger storage capacities to be shoved into the same form factors that we use today. Toshiba will begin output of NAND chips with circuitry widths in the upper 20 nanometre range soon, while production of chips with circuitry widths in the lower 20 nanometres is slated to start as early as 2012.

Samsung pioneers 20-nm NAND flash memory technology



April 19, 2010, By Thomas Antony in Storage



Right on the heels of Toshiba's announcement to start on sub-25nm flash memory, Samsung today announced the industry's first production of 20 nanometer class NAND chips for use in SD cards. Samsung's 20nm MLC 32-gigabit NAND chips are sampling now for use in embedded storage and SD memory cards ranging from 4GB to 64GB. This is a significant step forward for Samsung who started its 30nm production just one year ago. The new class of memory chips will allow for higher-density in storage, lower manufacturing costs and 50% higher productivity than 30nm technology.

POSTECH

- 8/48 -

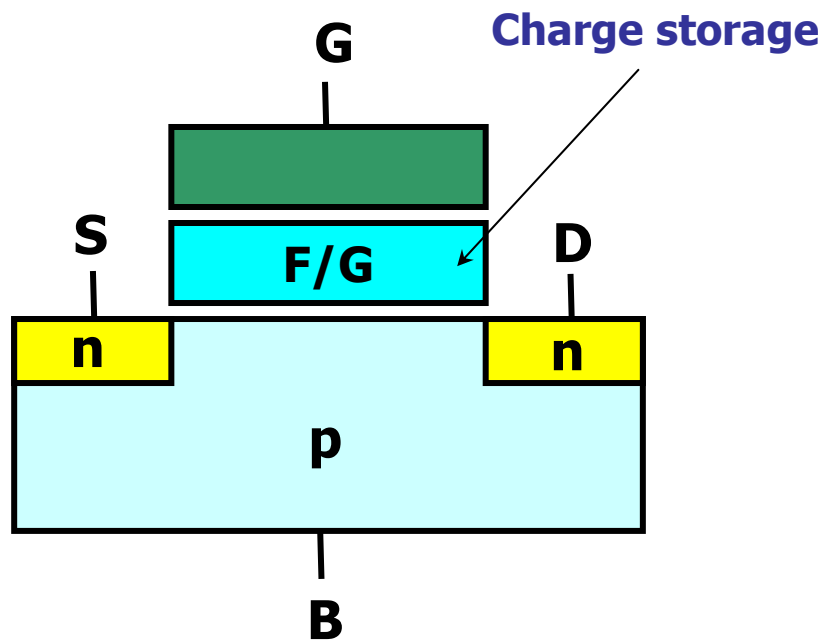


Flash memory 101

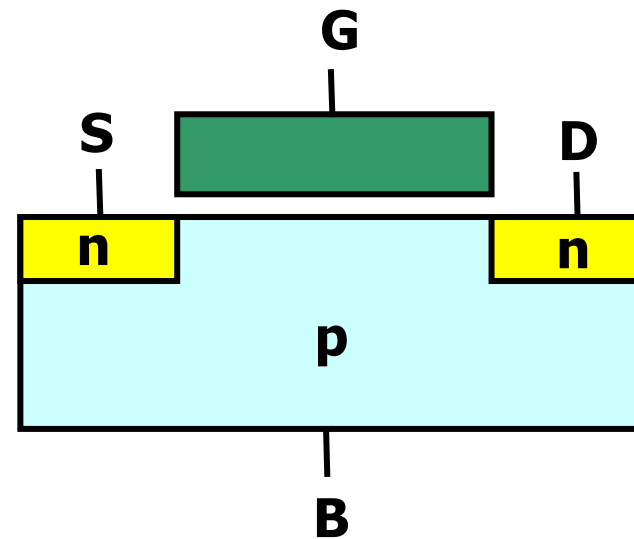


Flash memory cell vs. MOSFET

- Flash cell has a charge storage layer such that V_{th} of a cell can be changed → memorize information



Flash cell

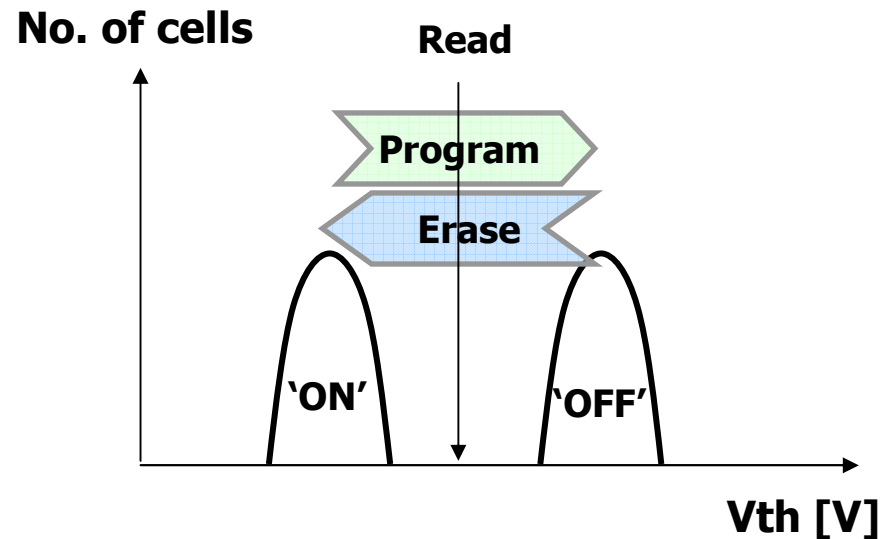


MOSFET

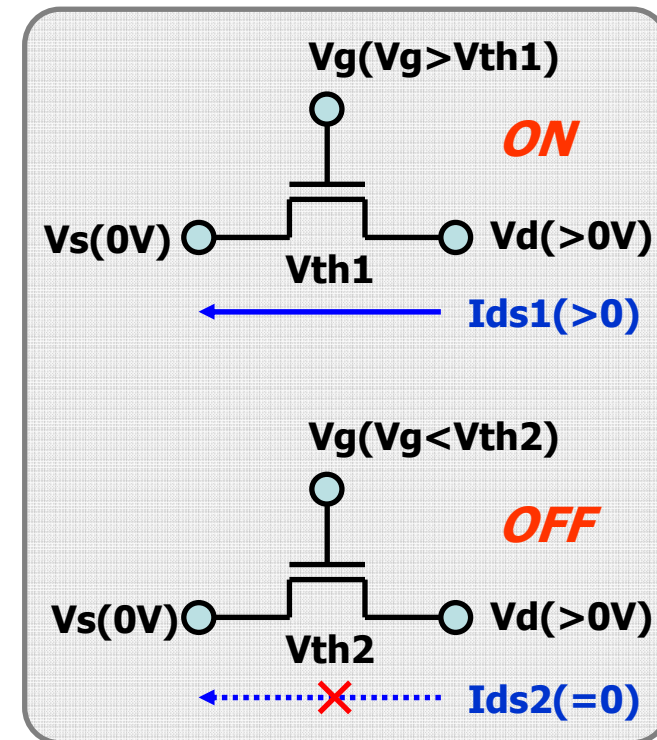
Flash memory operation

- Write & read binary data to a flash cell

- data '0' → 'OFF' state (program)
- data '1' → 'ON' state (erase)

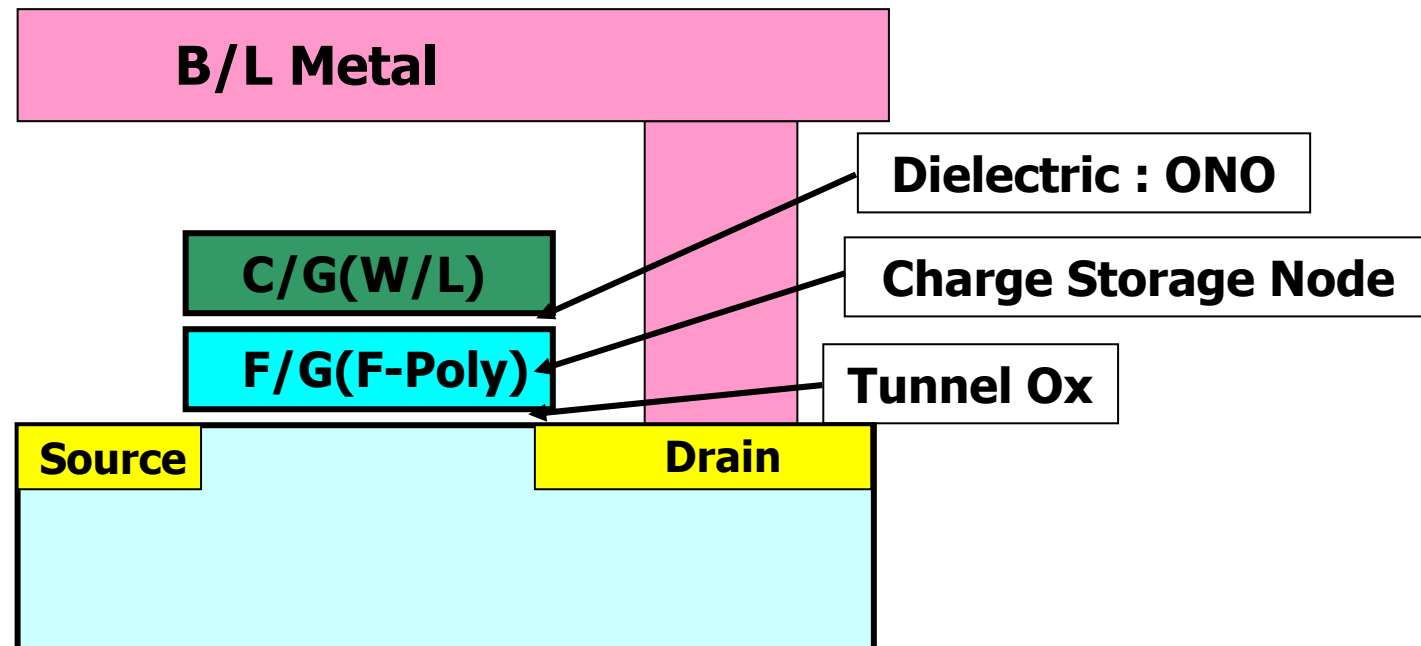


MOS Transistor



Flash memory cell structure

- Cell V_{th} changes depending on the amount of F/G charge
- electrons can be injected(ejected) into(out of) the F/G through T_{ox} with electric field across T_{ox}



NAND vs. NOR

Truth table

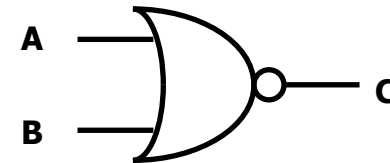
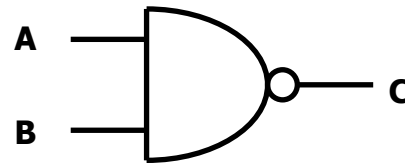
NAND

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

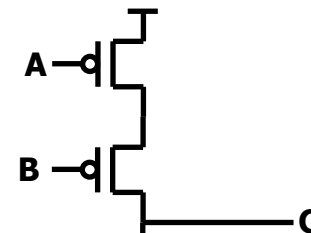
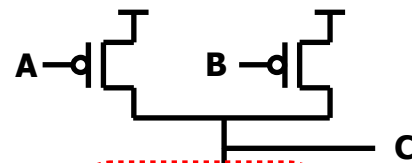
NOR

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

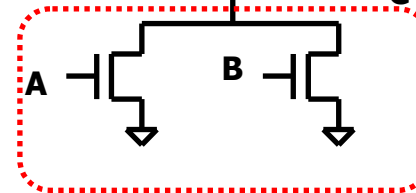
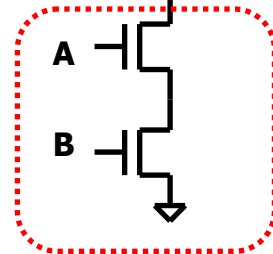
Logic gate



circuit



Flash



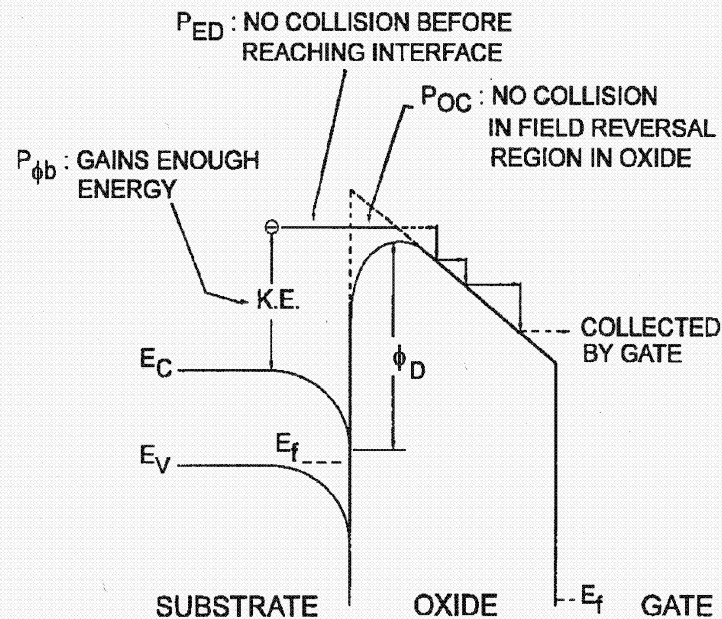
Features : NAND vs. NOR

	NAND	NOR
Cell Array & Size	Word line Unit Cell Source line $5F^2$ $2F$ $2.5F$	Word line Unit Cell Source line $10F^2$ $2F$ $5F$ Bit line Contact
Cross-section		
Features	Small Cell Size, High Density Low Power & Good Endurance → <i>Mass Storage</i>	Large Cell Current, Fast Random Access → <i>Code Storage</i>



Write methods in Flash memory

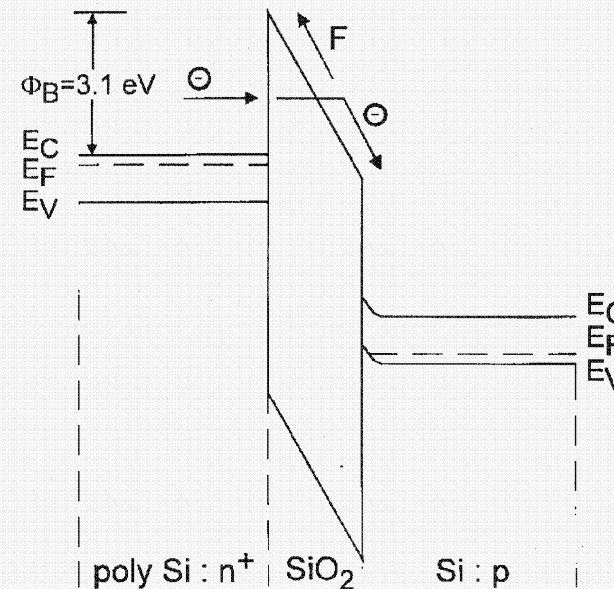
Hot Electron Injection



Program in NOR Flash

Impact ionization at drain side

Fowler-Nordheim Tunneling

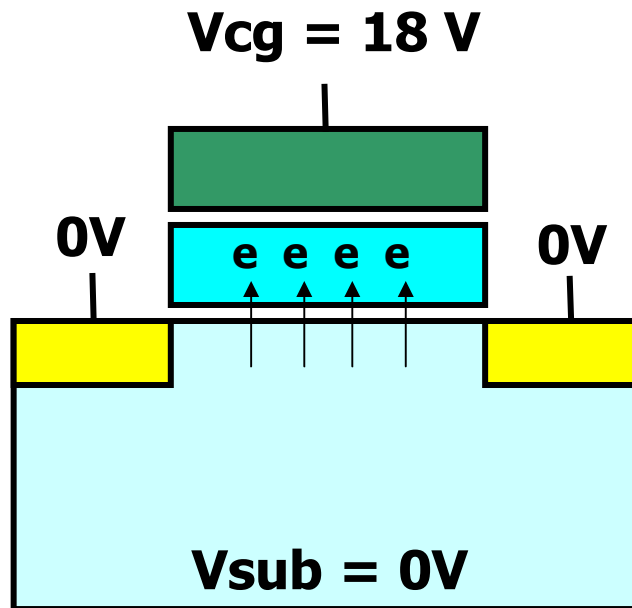


Program/Erase in NAND Flash

Erase in NOR Flash

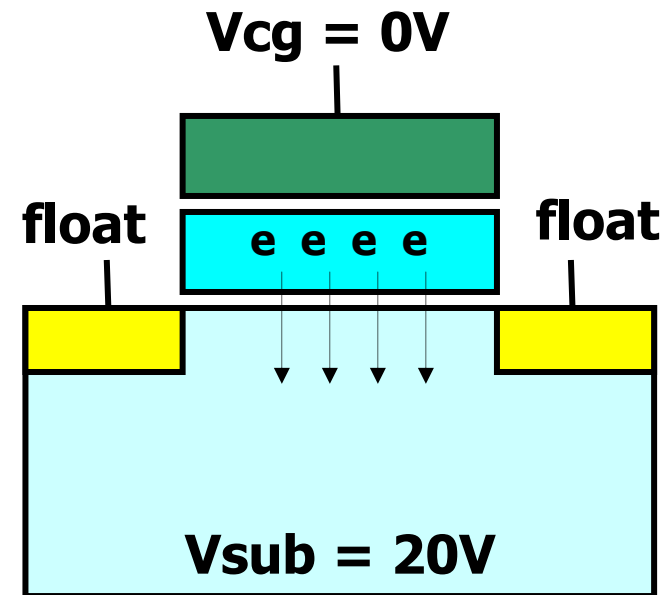


NAND Flash : Program & Erase



**Program
F-N Tunneling**

**Off cell
(Solid-0)**

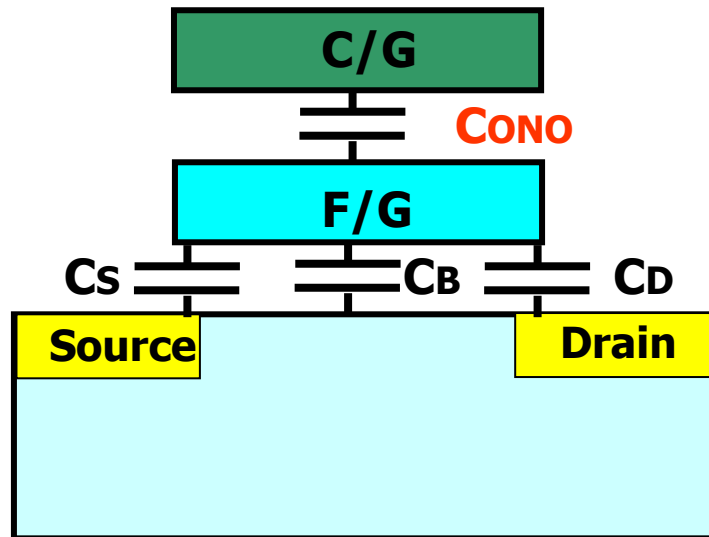


**Erase
F-N Tunneling**

**On cell
(Solid-1)**



Coupling ratio



$$V_{fg} = V_{cg} \times \alpha_{cg}$$

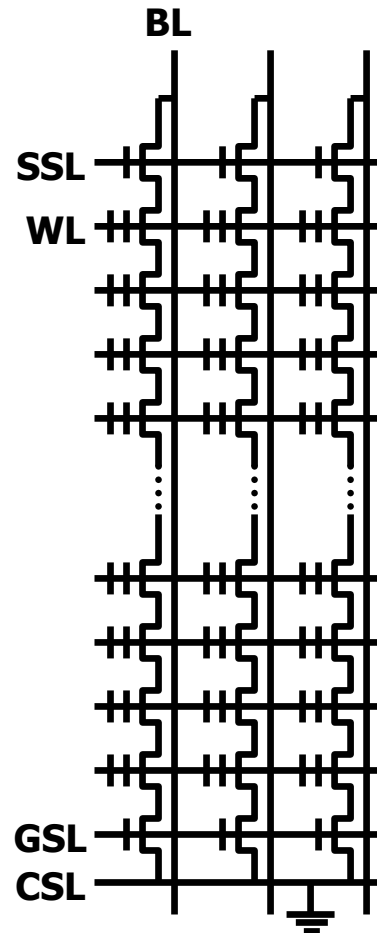
$$\alpha_{cg} = \frac{C_{ONO}}{(C_D + C_S + C_B + C_{ONO})}$$

α_{cg} : Coupling Ratio

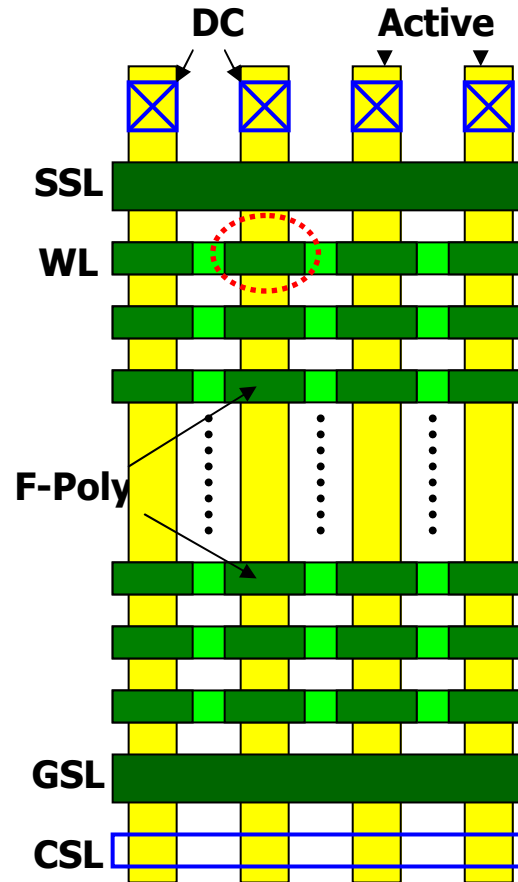
(From $Q=CV$ and
Charge Conservation Law)

For fast programming, high V_{fg} is required,
i.e. either high V_{cg} or large α_{cg}

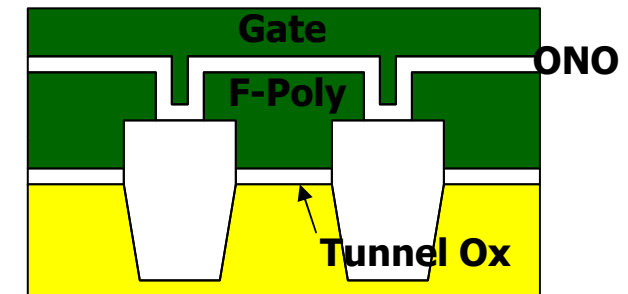
NAND Flash cell structure



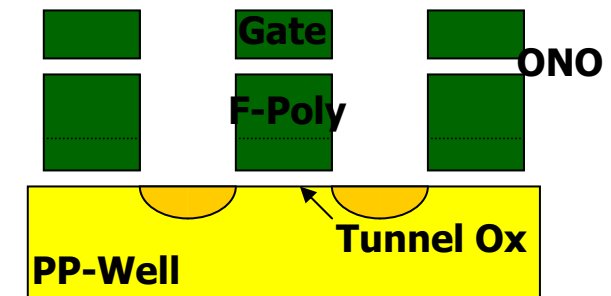
Schematic



Top View



WL Direction

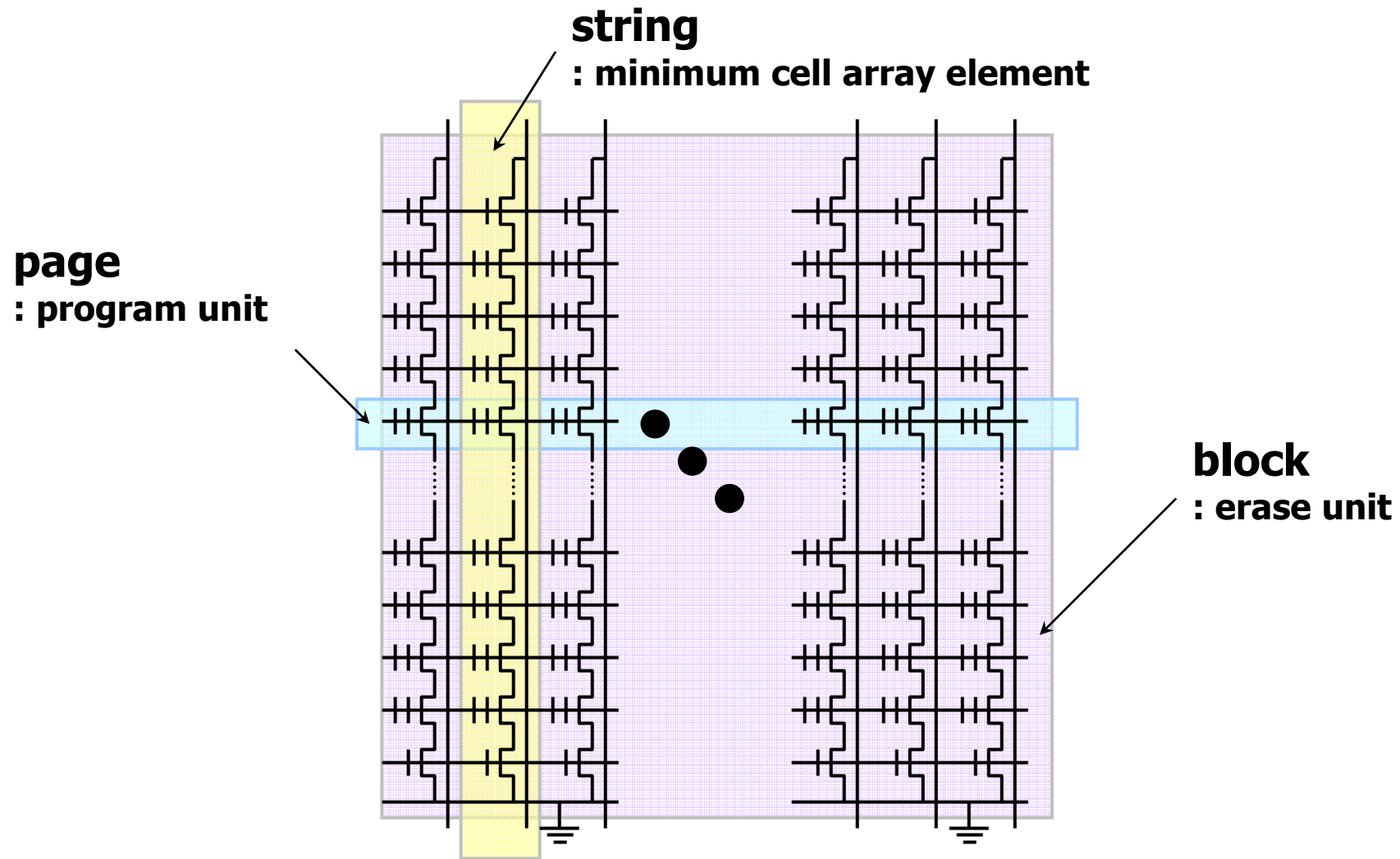


BL Direction

Vertical View

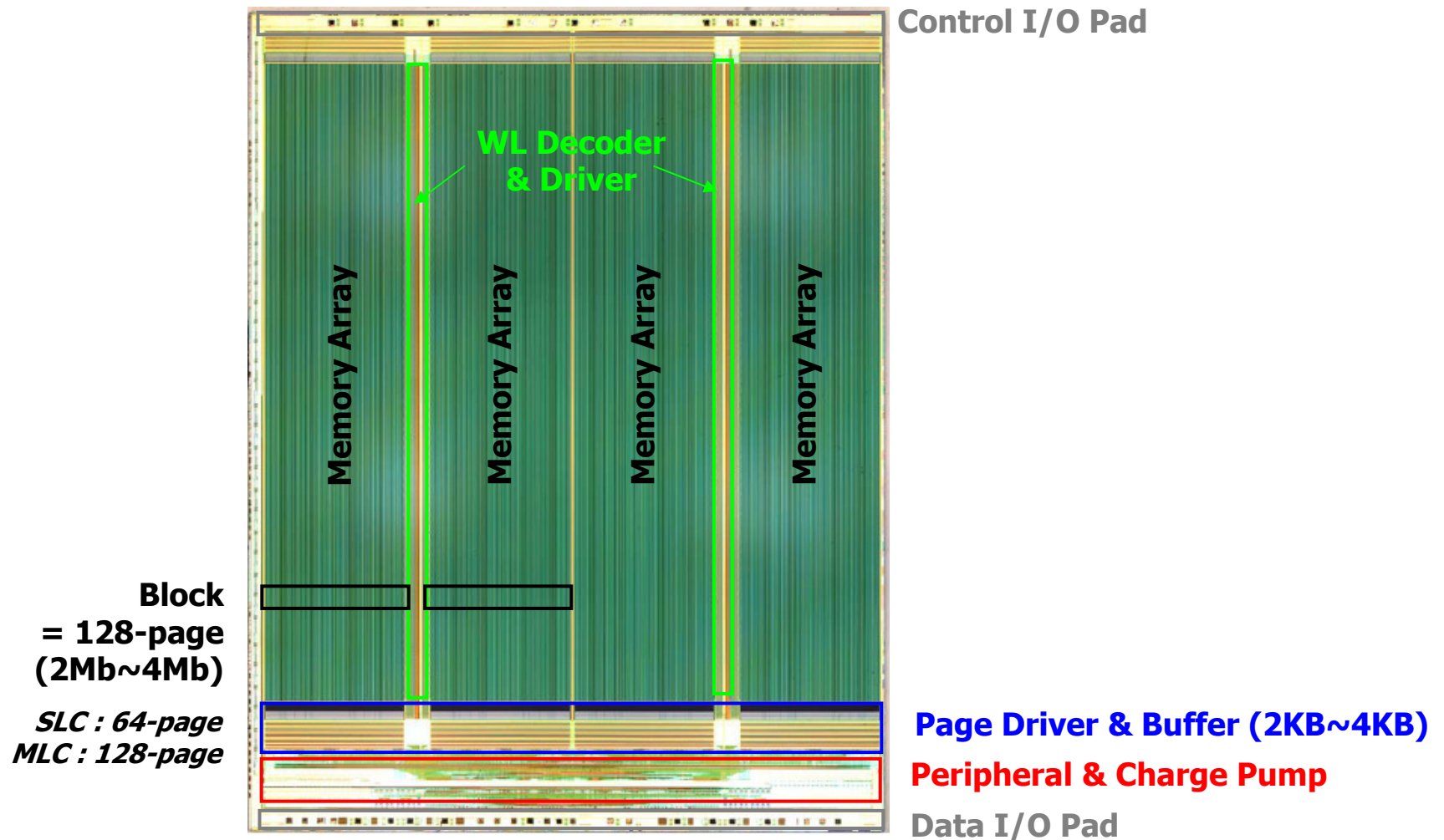


Terms in NAND Flash – string, page, block

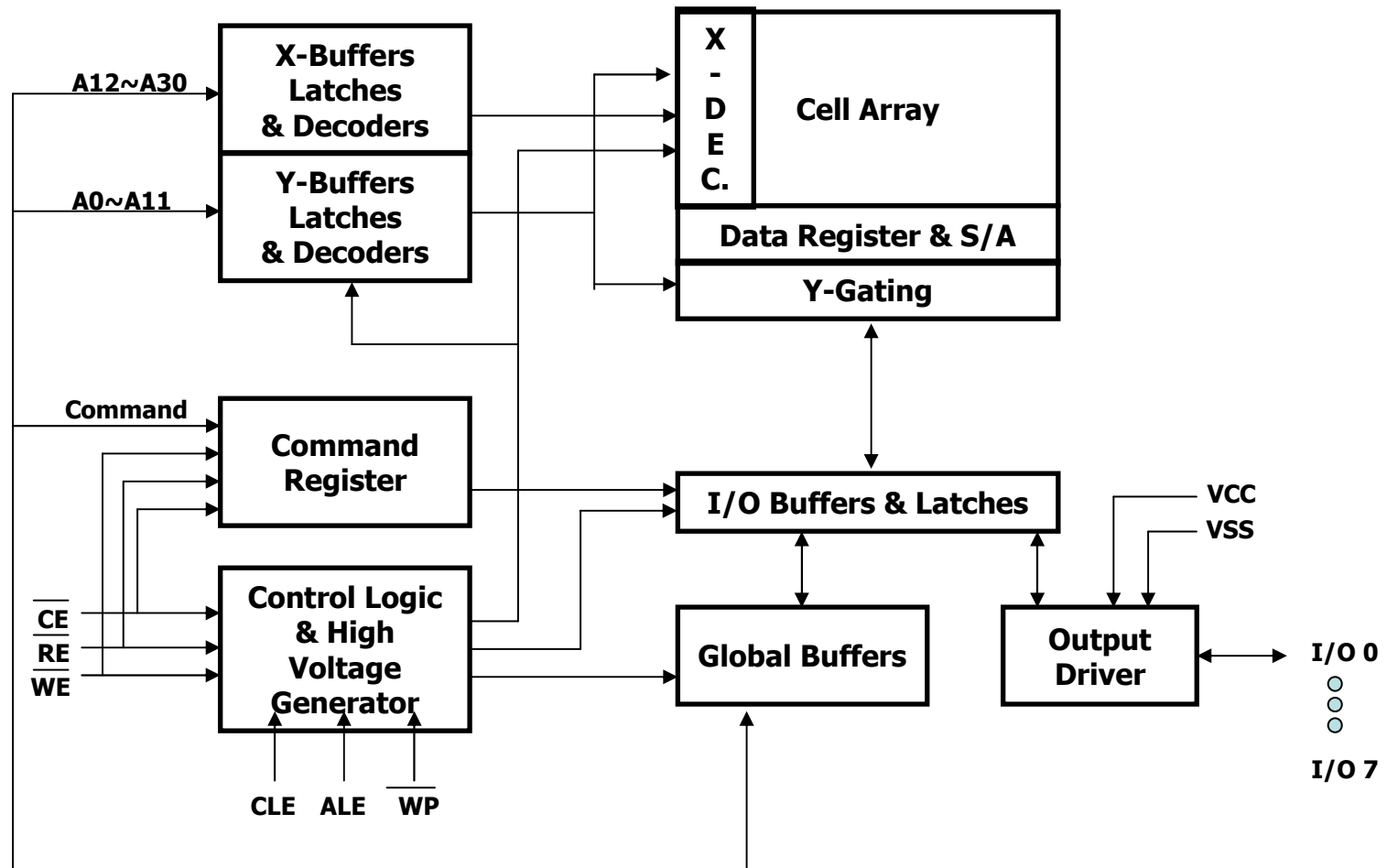


NAND Flash chip architecture

- High density & simple architecture (cell efficiency > 65%)



Functional block diagram of NAND Flash



Basic operations

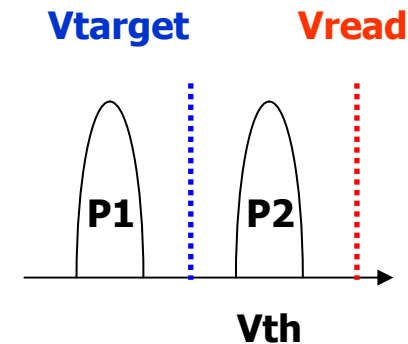
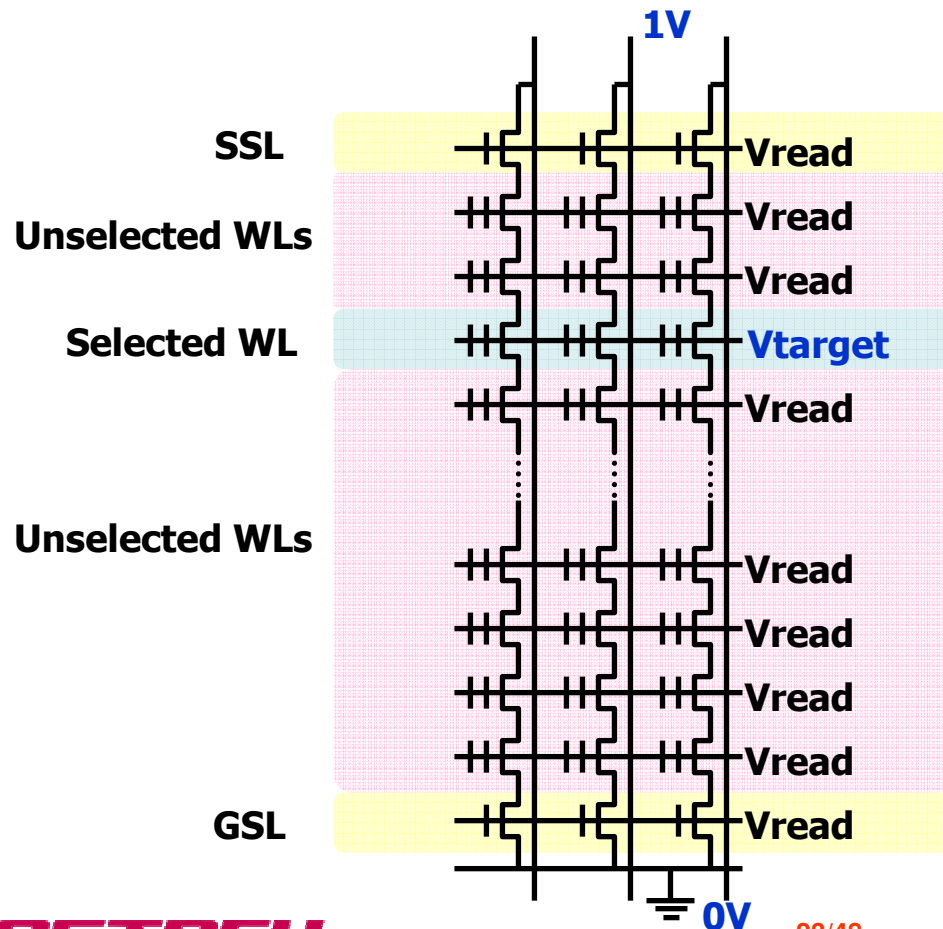
NAND Flash



Read operation

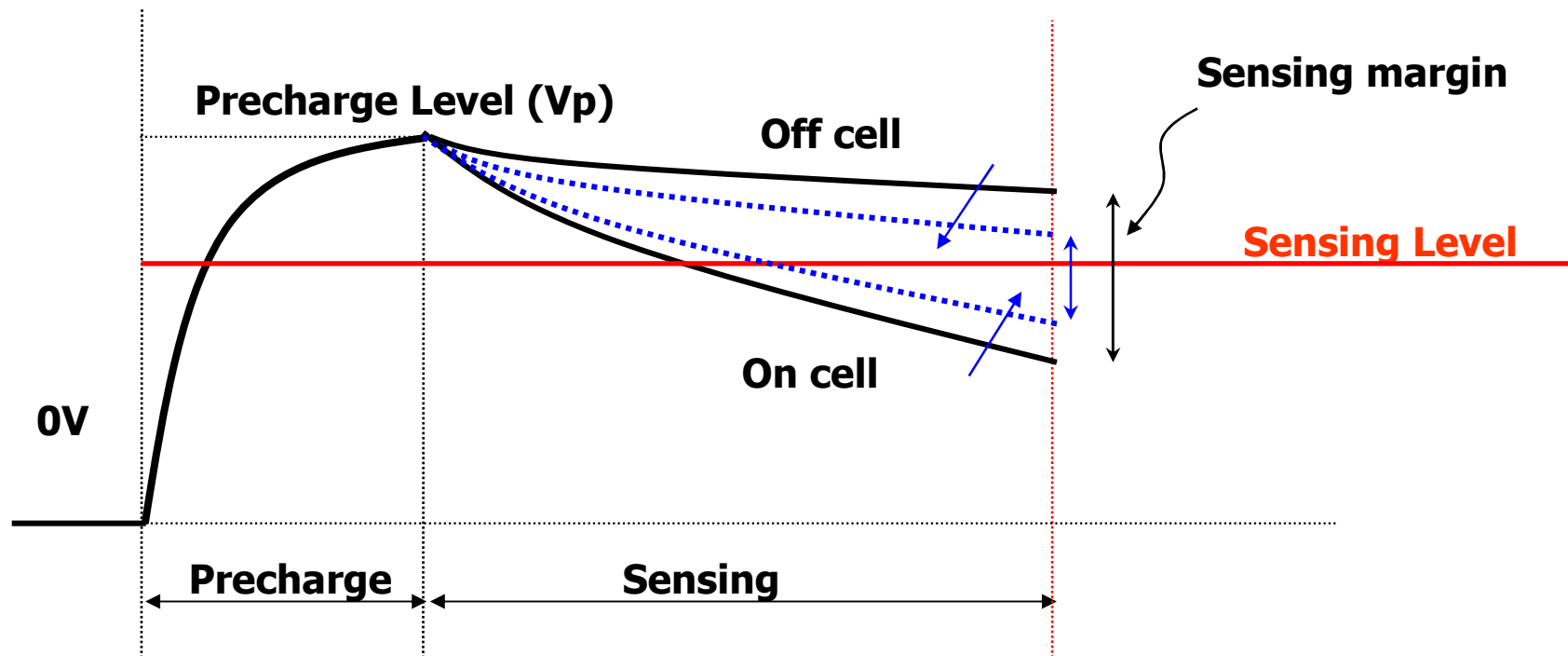
● Bias condition

- selected WL : Target voltage (V_{target})
- unselected WLs : high enough to conduct all cells in a string



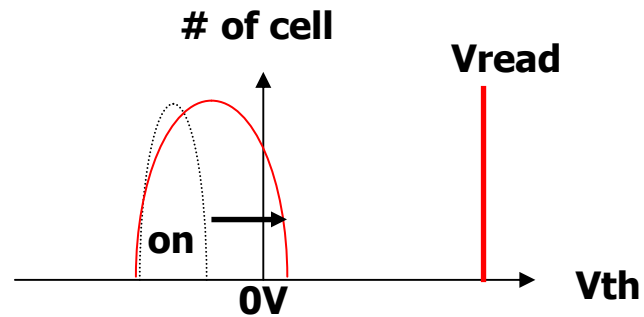
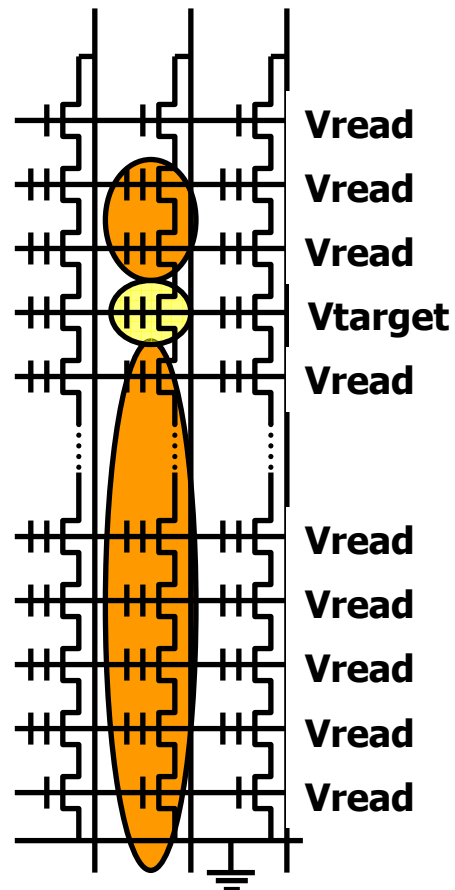
Sensing margin

- The ratio of On cell & Off cell current



Read disturbance

- Increasing V_{read} → soft program occurs in the unselected cell of selected string

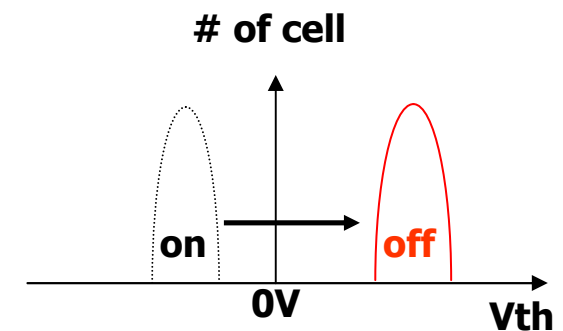
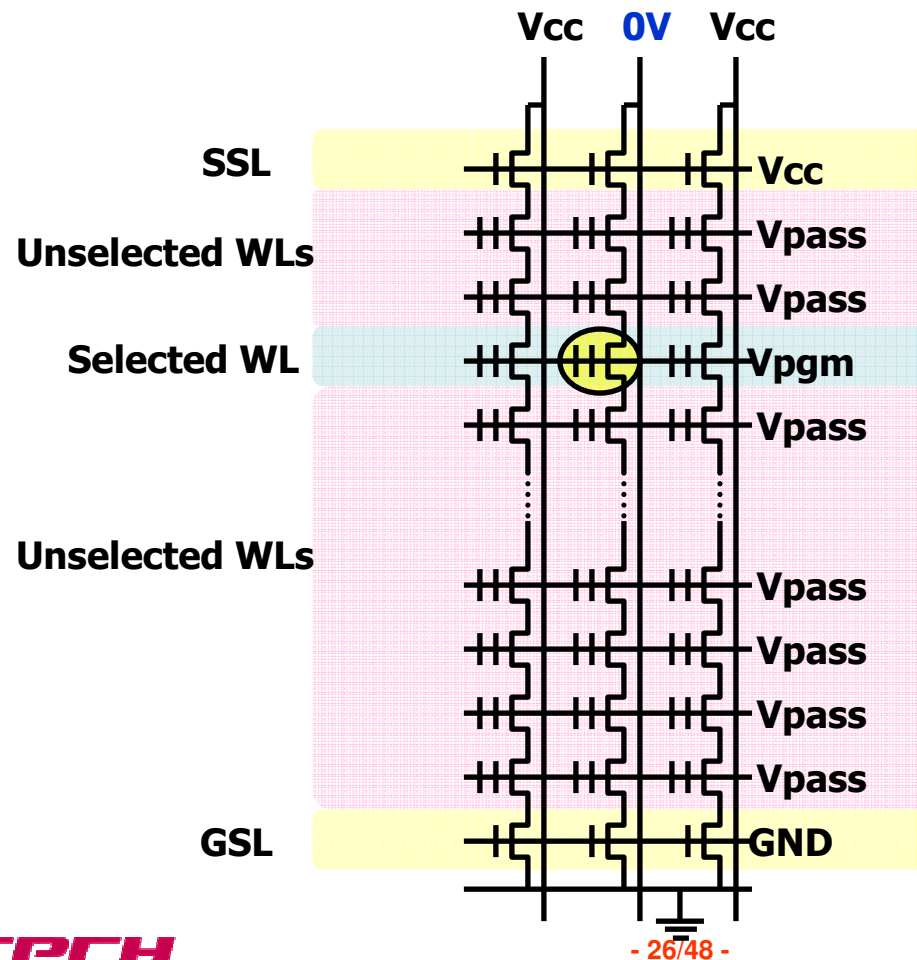


On cell moves to off cell

Program operation

● Bias condition

- selected WL : Program voltage (V_{pgm})
- unselected WLs : Pass voltage (V_{pass})

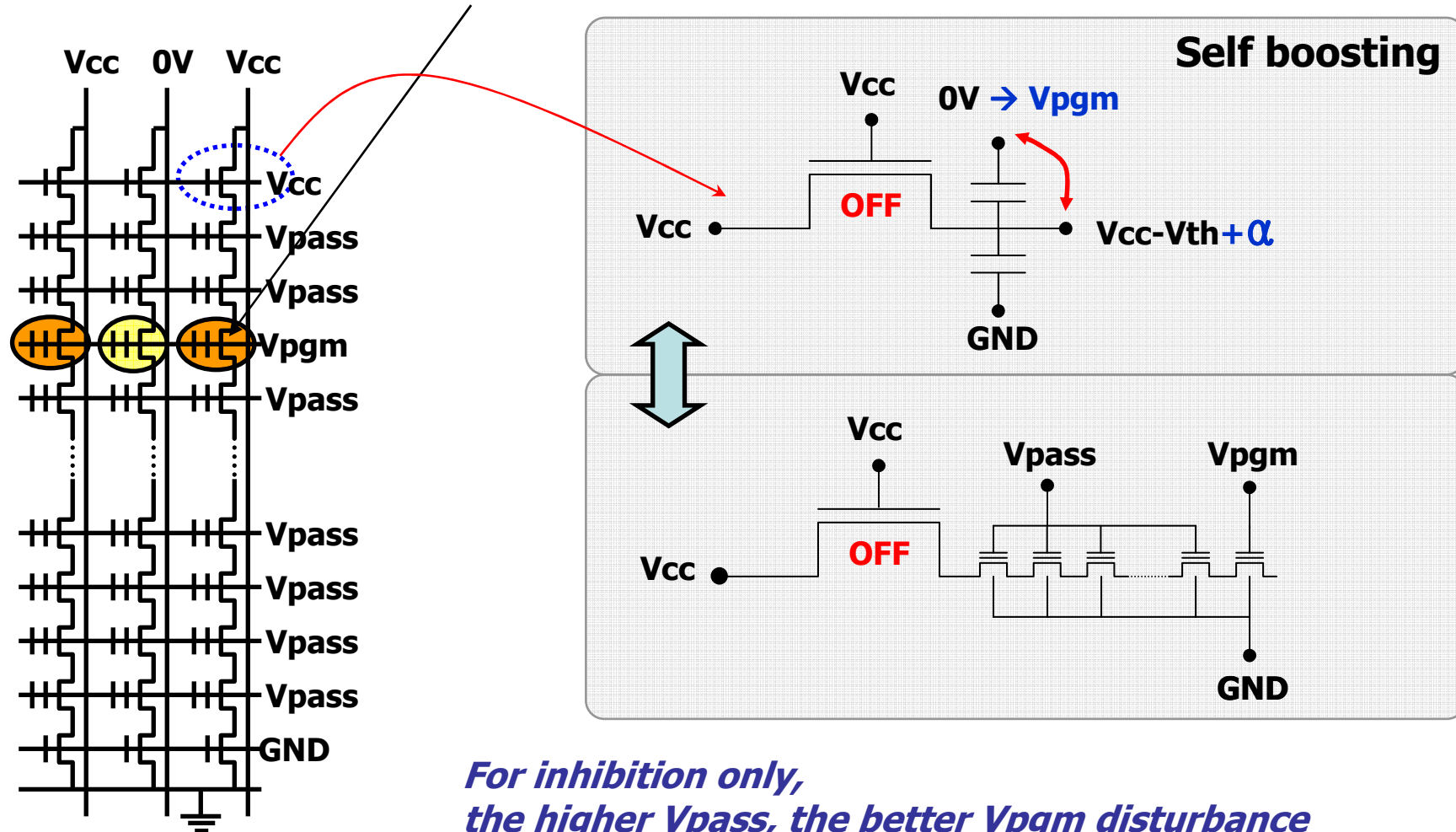


on cell becomes off cell



Program inhibition

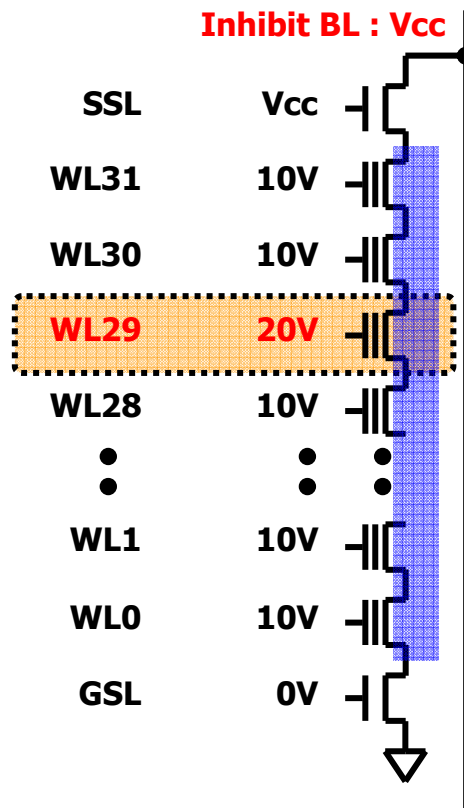
Should not be programmed (= program inhibited)



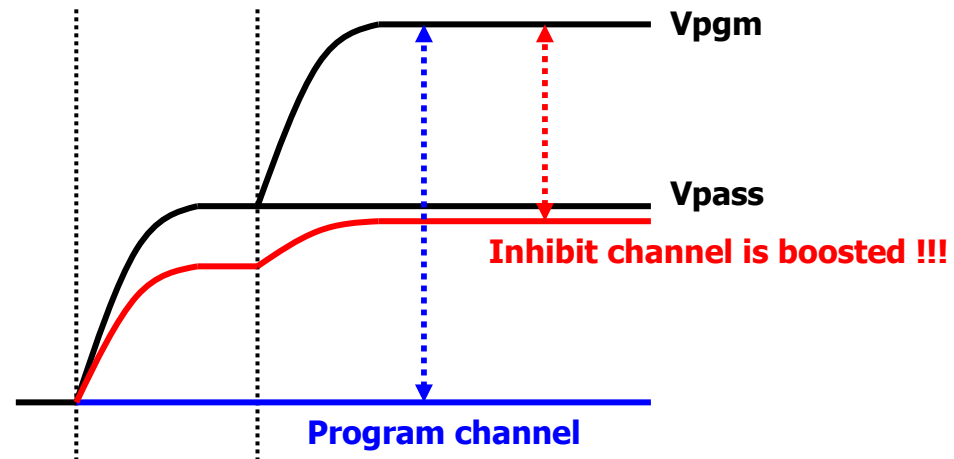
*For inhibition only,
the higher V_{pass} , the better V_{pgm} disturbance
But, higher V_{pass} causes more V_{pass} disturbance*



Self boosting



Bias sequence for self boosting

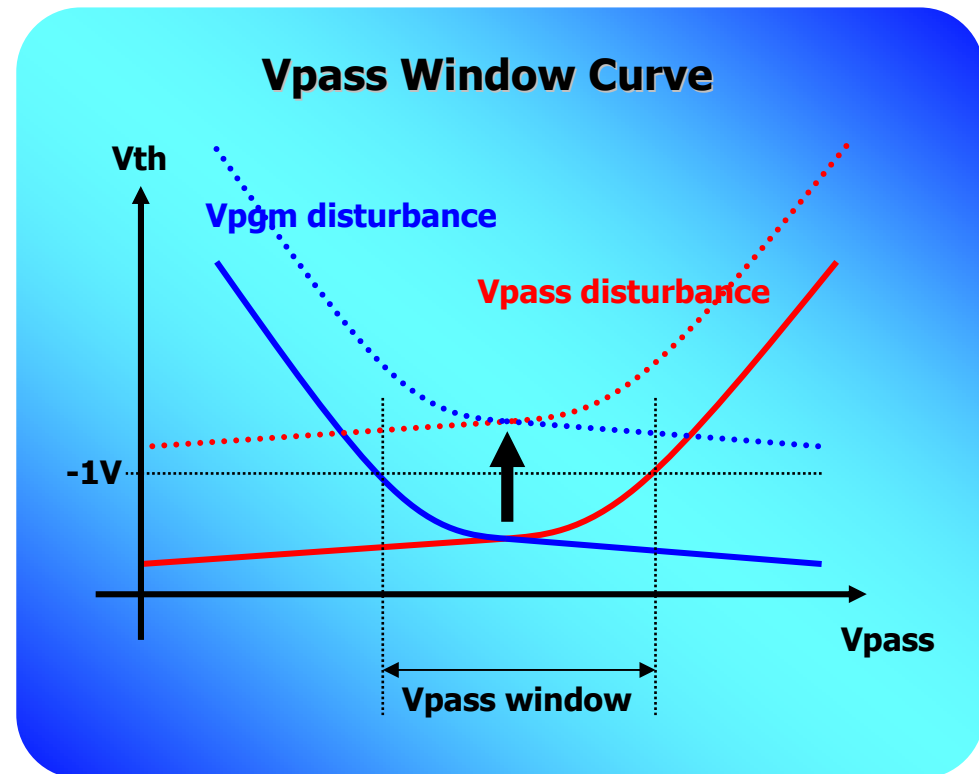
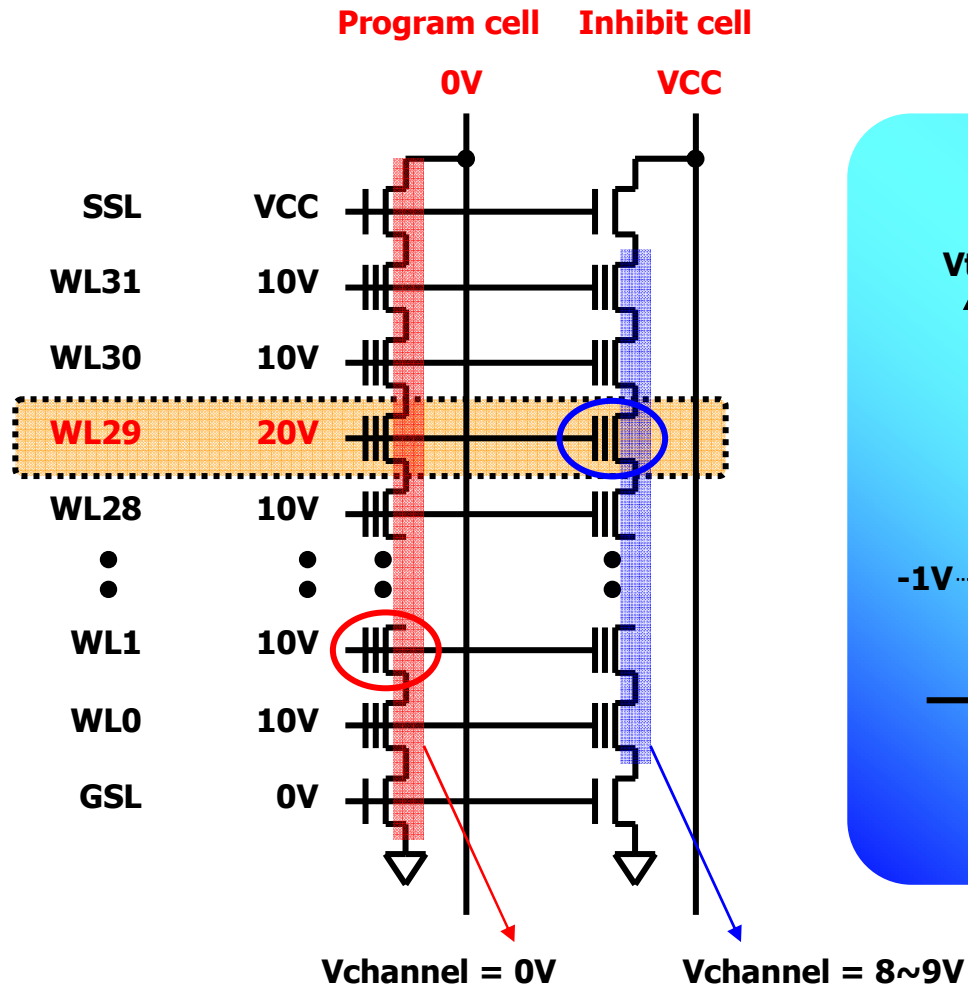


Channel potential strongly depends on the cell states in a string



Vpass window

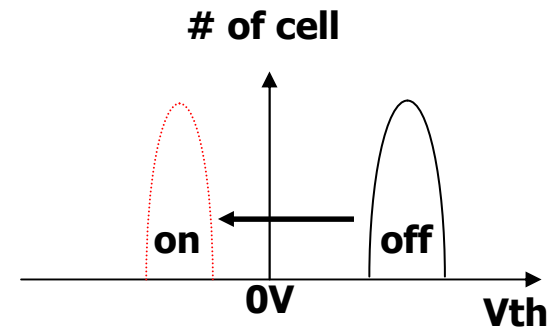
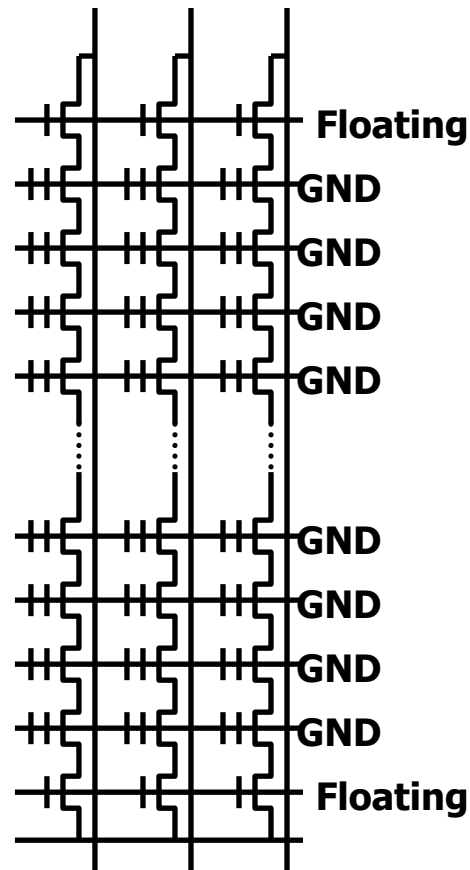
- “Optimal Vpass region” considering both Vpgm and Vpass disturbances at the same time



Erase operation

● Bias condition

- all WLs in the selected Block : 0V
- GSL/SSL : Floating
- Bulk : Vera

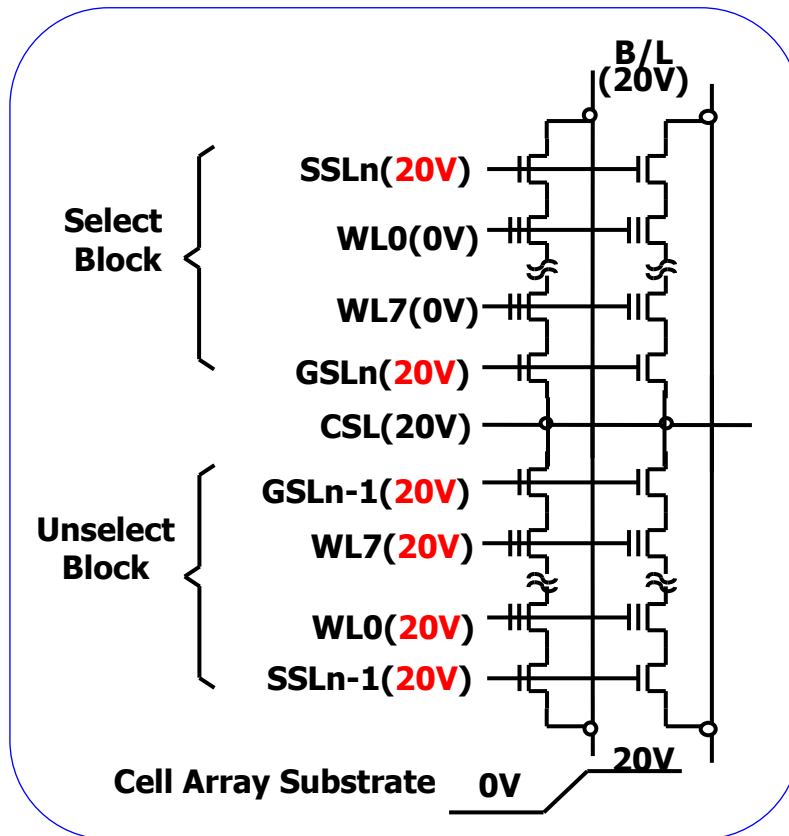


off cell becomes on cell

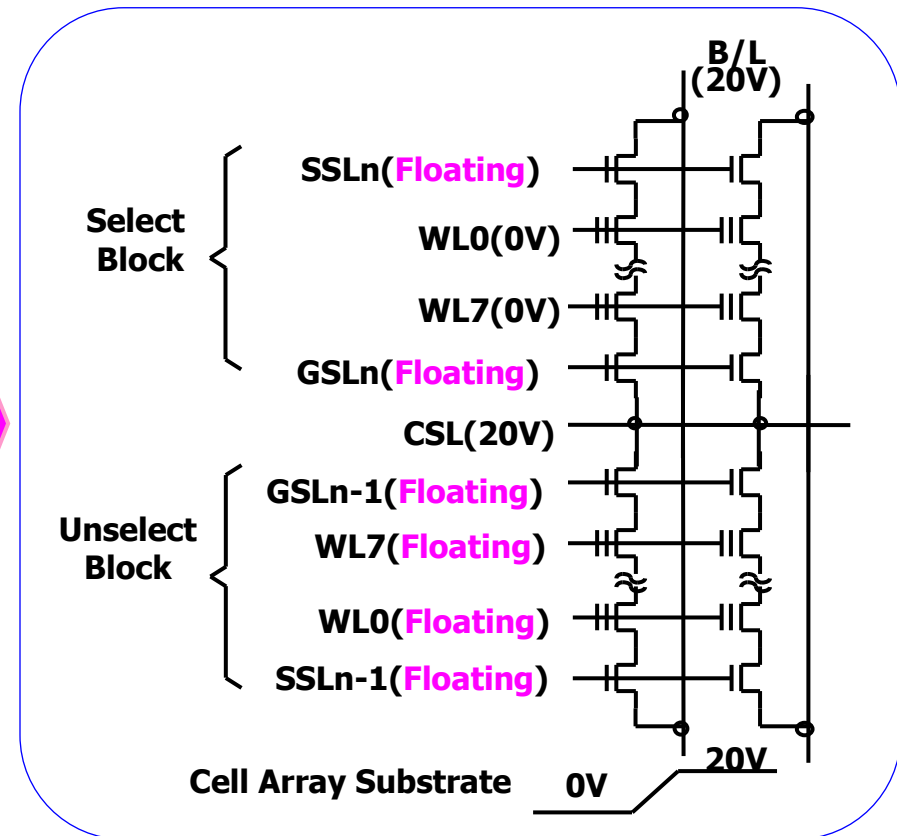
Erase disturbance

- "Self boosting" can be used

Old Method

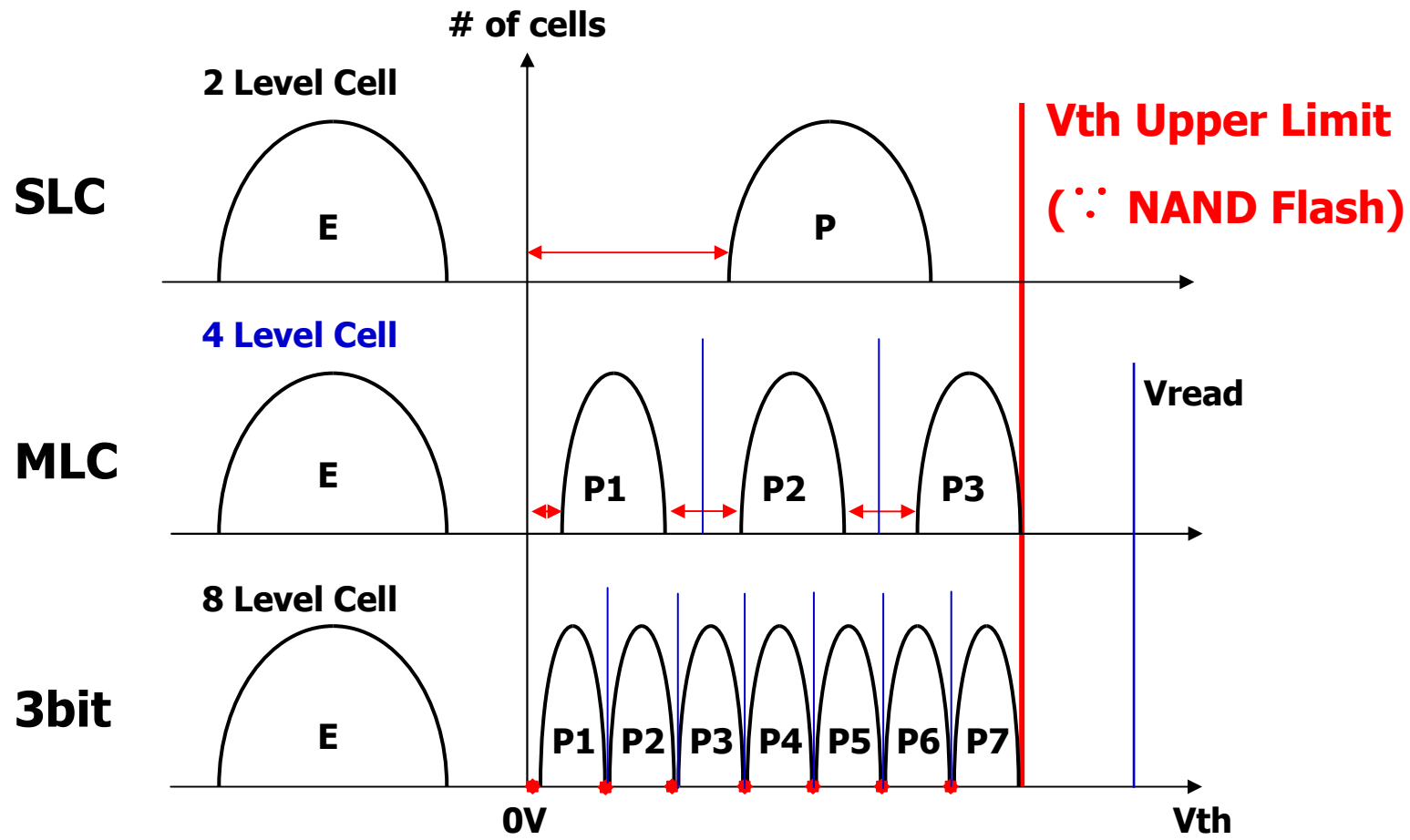


New Method



Cell Vth distribution

- Cell Vth width requirement
 - 1bit/cell vs. 2bit/cell vs. 3bit/cell

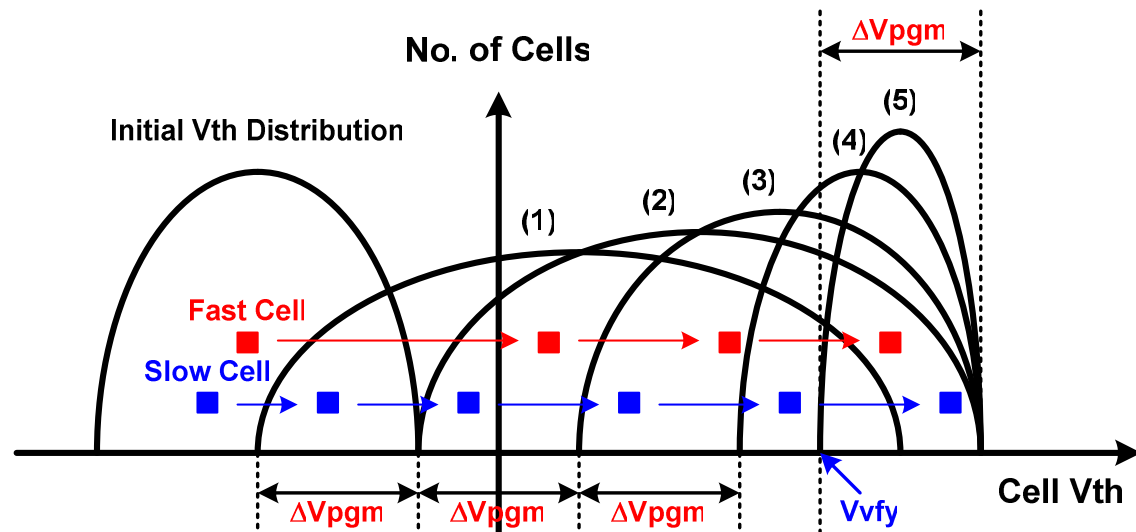
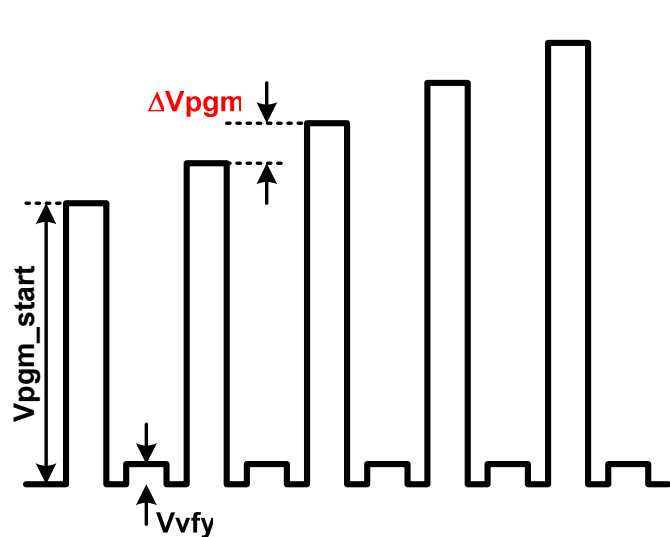
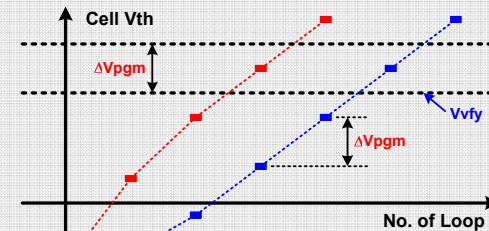


Cell Vth width control

● Incremental Step Pulse Program (ISPP)

- programmed state Vth width can be controlled
- narrower width requires more program loop

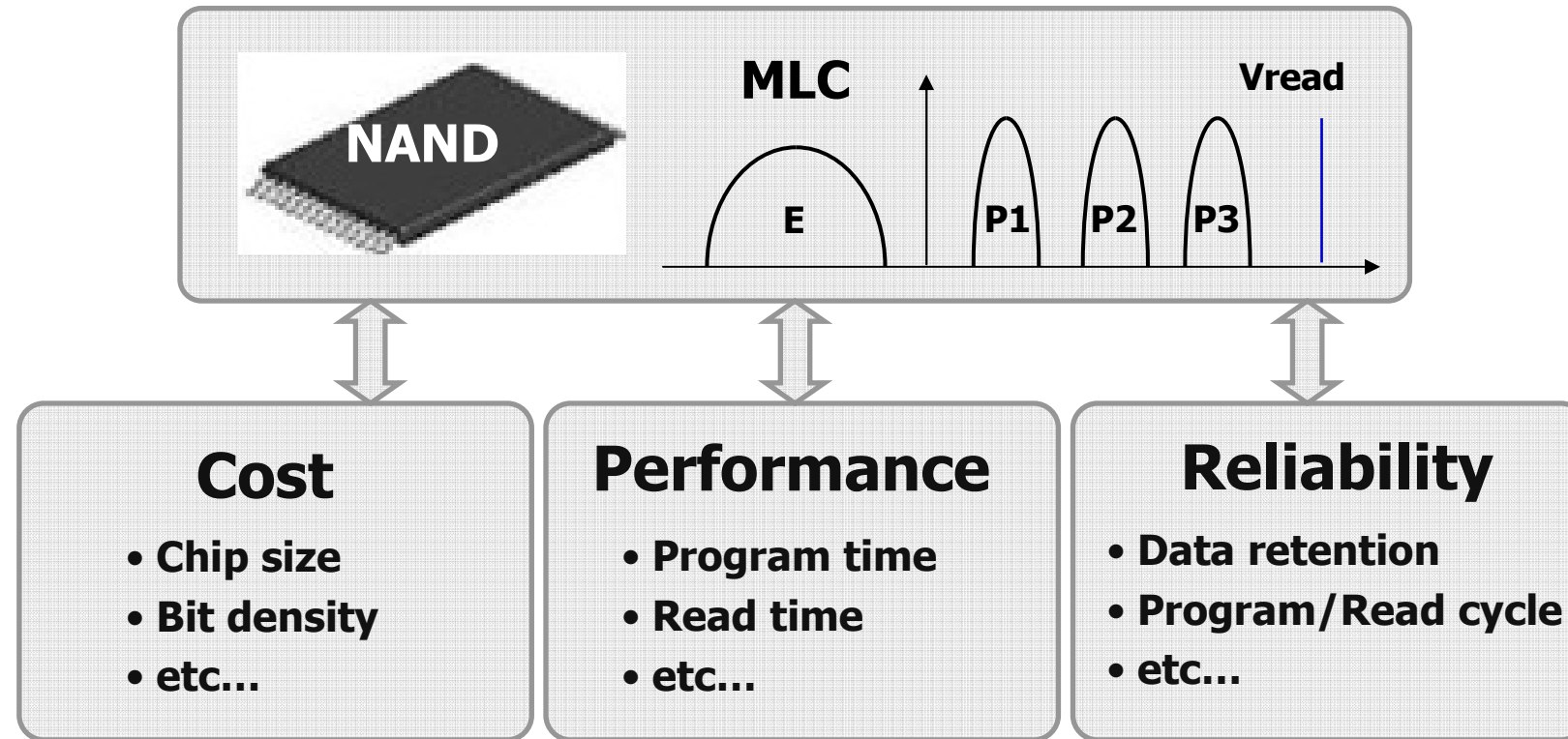
Change of cell Vth : slow vs. fast cell



Current issues & approaches



Criteria for a NAND Flash device

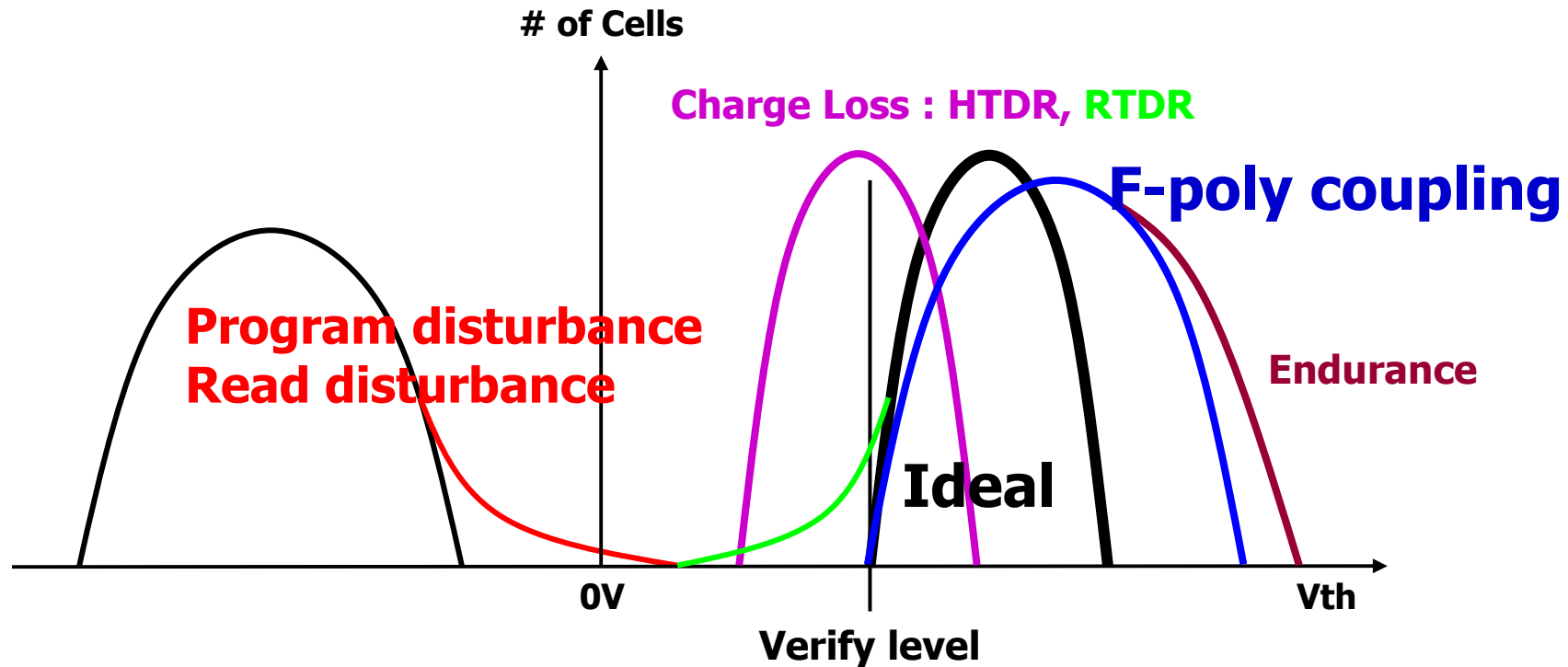


All these are strongly dependent on each other and may become worse as "cell size shrinks down"



Details on programmed Vth

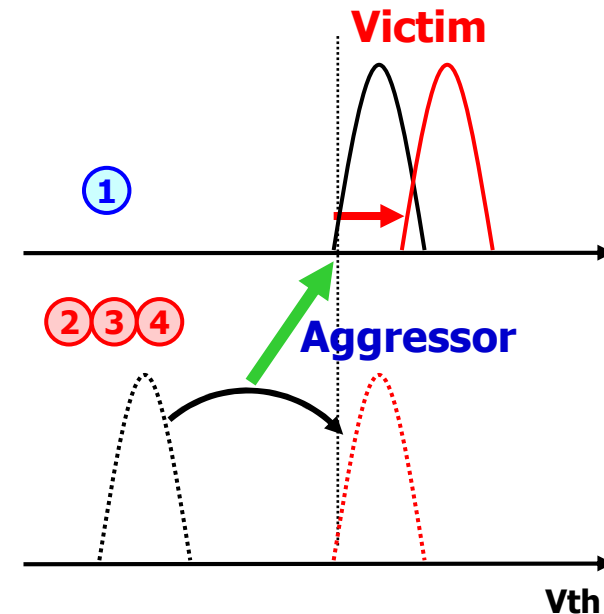
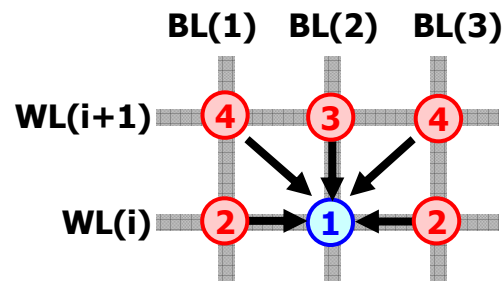
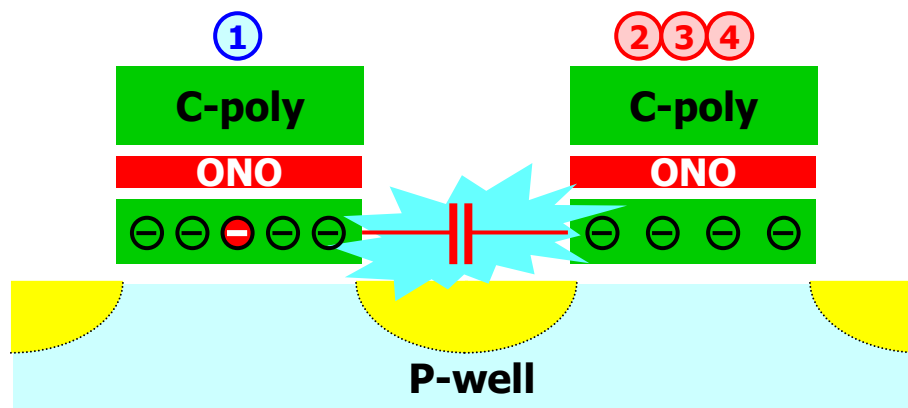
- Factors which affect to programmed Vth width
 - Ideal : ΔV_{pgm} during ISPP program
 - Noise : F-poly coupling, CSL noise, Back pattern dependency
 - Reliability : Endurance, program/read disturbance, charge loss



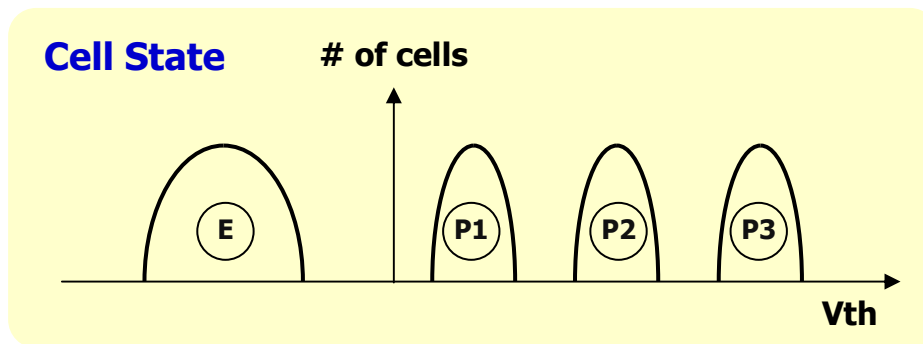
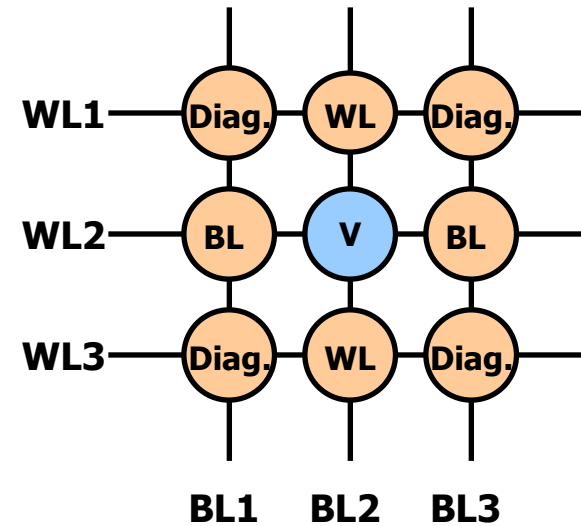
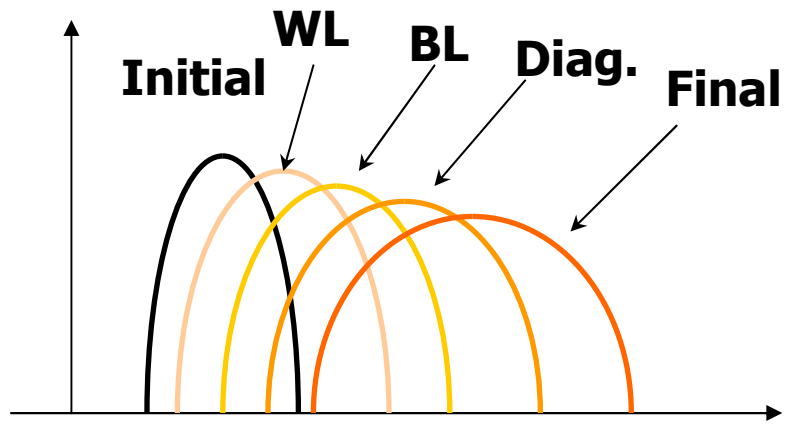
Neighborhood interference

- F-poly coupling noise

- Cell V_{th} can be raised as neighboring cells are programmed

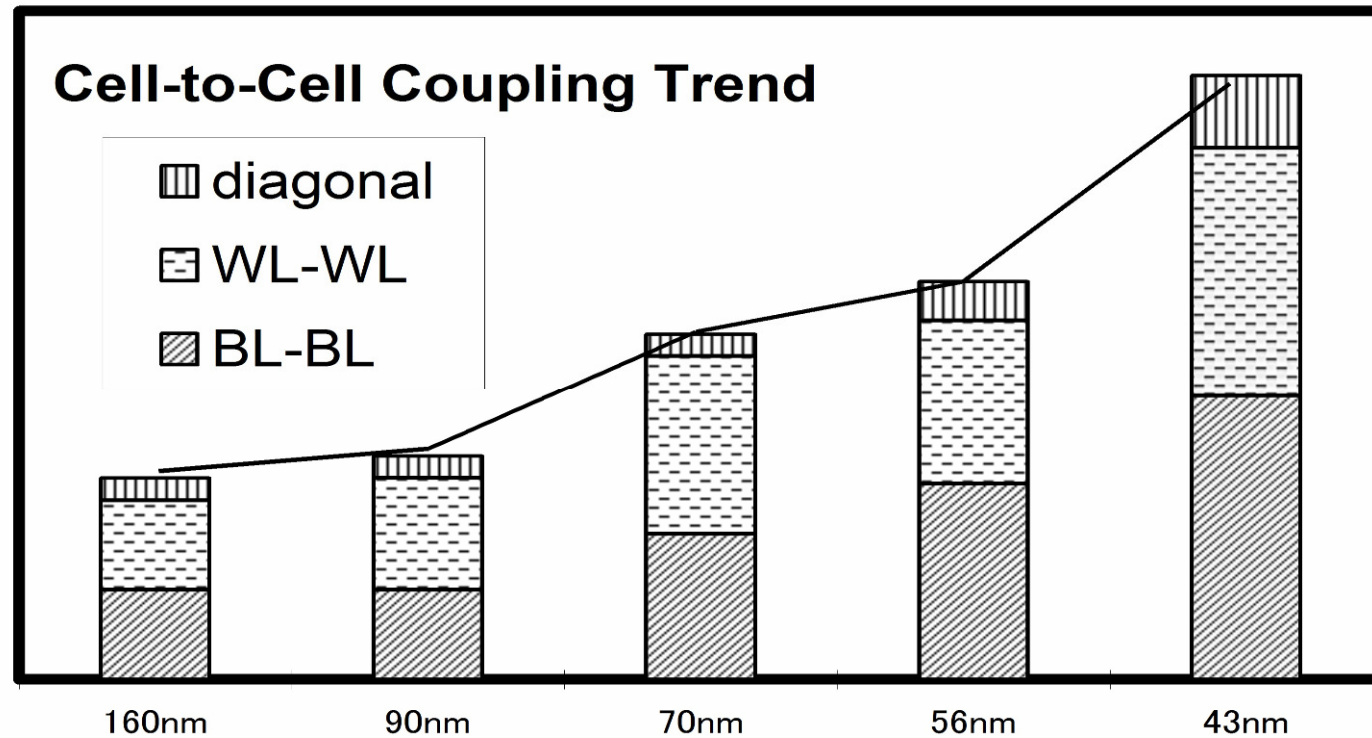


Cell Vth vs. F-poly coupling



Cell size vs. F-poly coupling

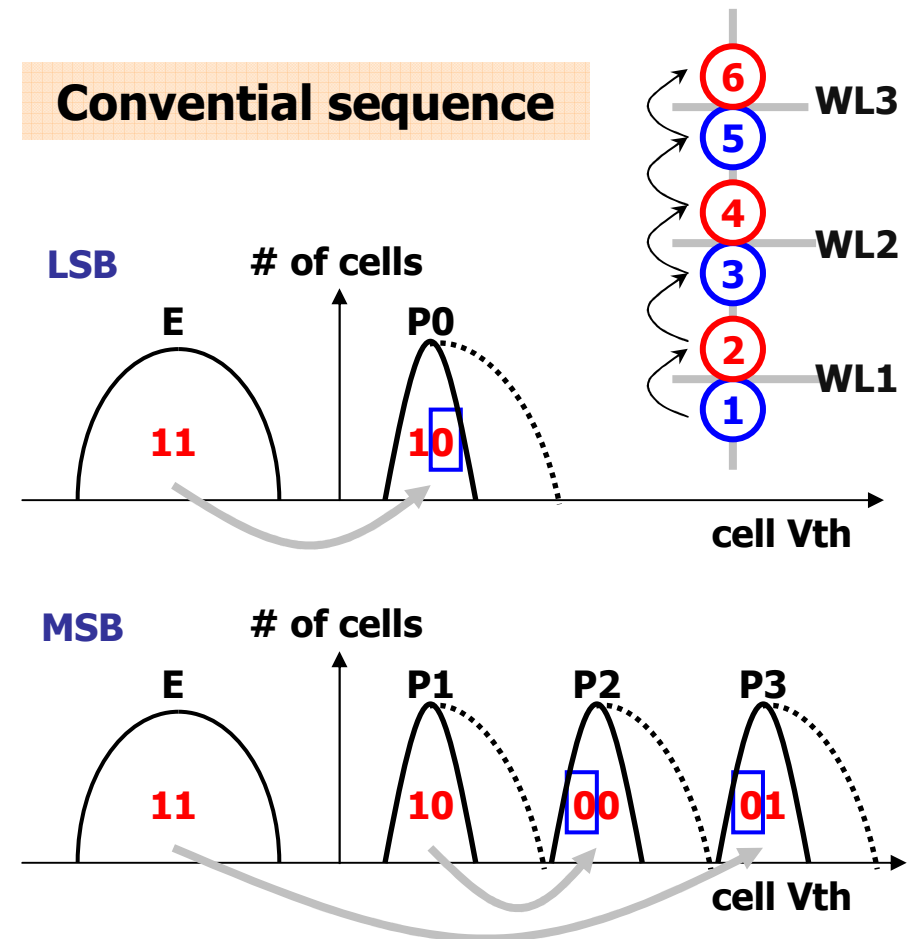
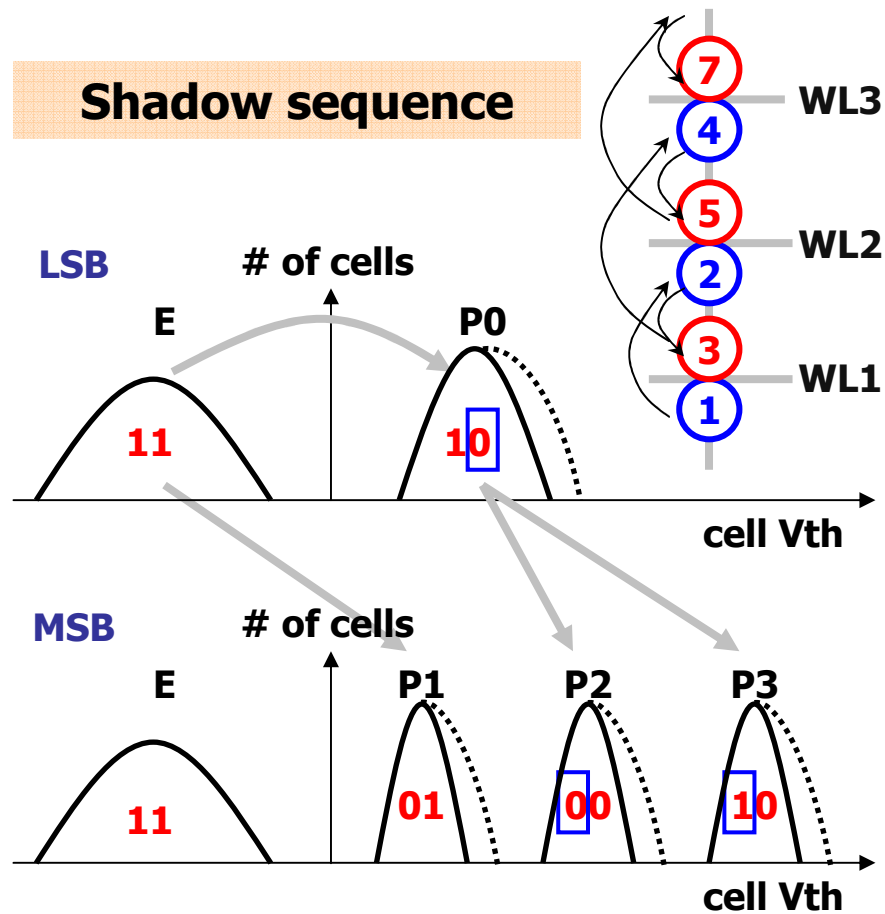
'09 ISSCC



F-poly coupling reduction – (I)

- Shadow program

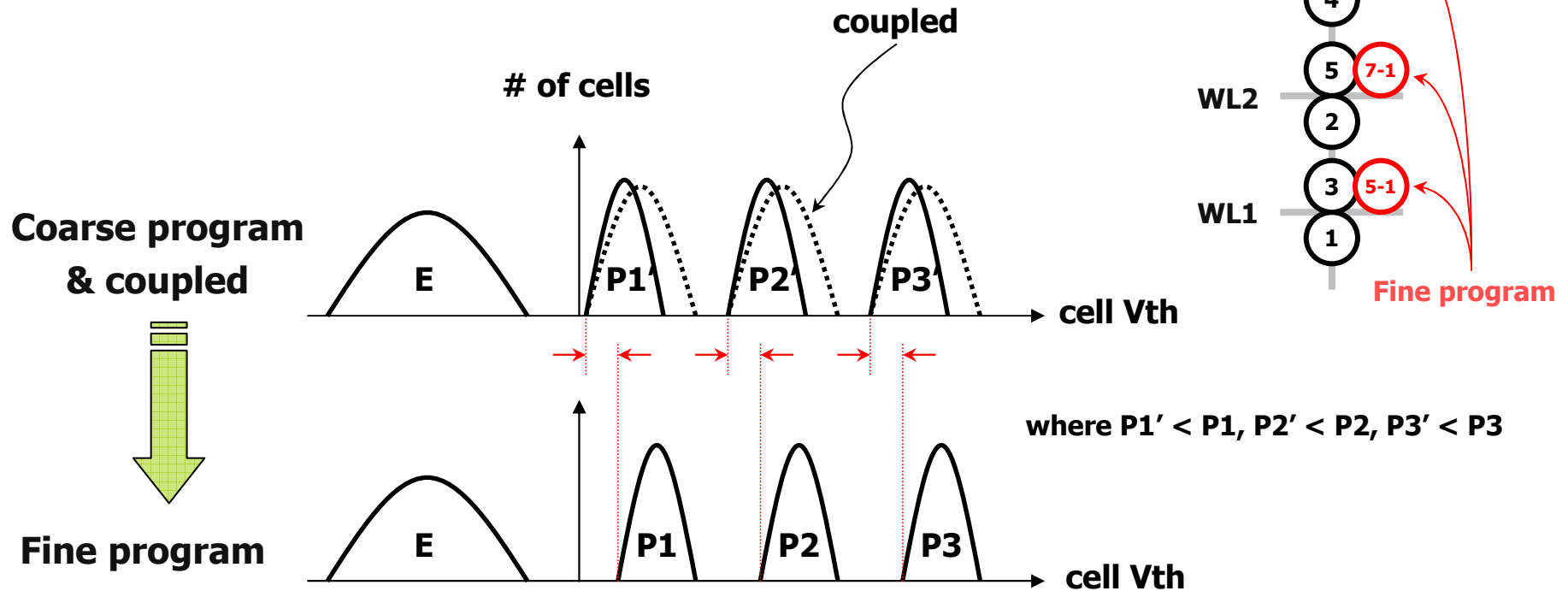
- Make final V_{th} after being coupled



F-poly coupling reduction – (II)

- Reprogram

- Make final V_{th} after being coupled



Program performance overhead due to "Fine program"

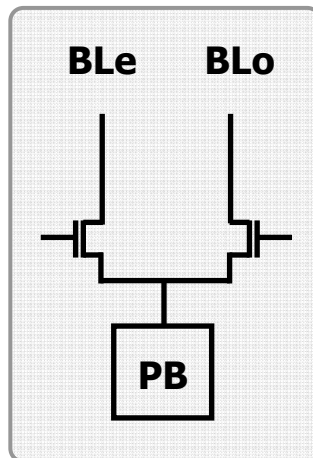


F-poly coupling reduction – (III)

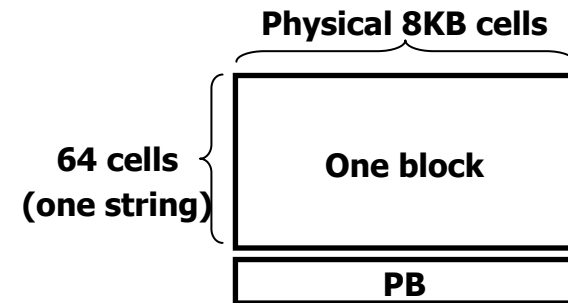
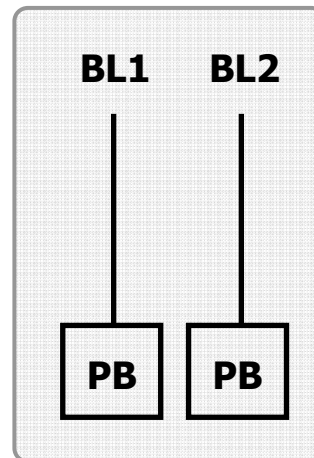
- **All-bit line (ABL) architecture**

- all cells in a WL are programmed at the same time
- in SBL, cells in even BL first, cells in odd BL next (BL-coupling exists)

SBL(Shielded-bit line)



ABL(All-bit line)



Features : SBL vs. ABL

	SBL	ABL
No. of PBs (page depth)	4KB	8KB
pages/block	256 pages	128 pages



In the near future



3D Flash memory

● Bit-Cost Scalable(BiCS) technology

2007 IEDM, Toshiba

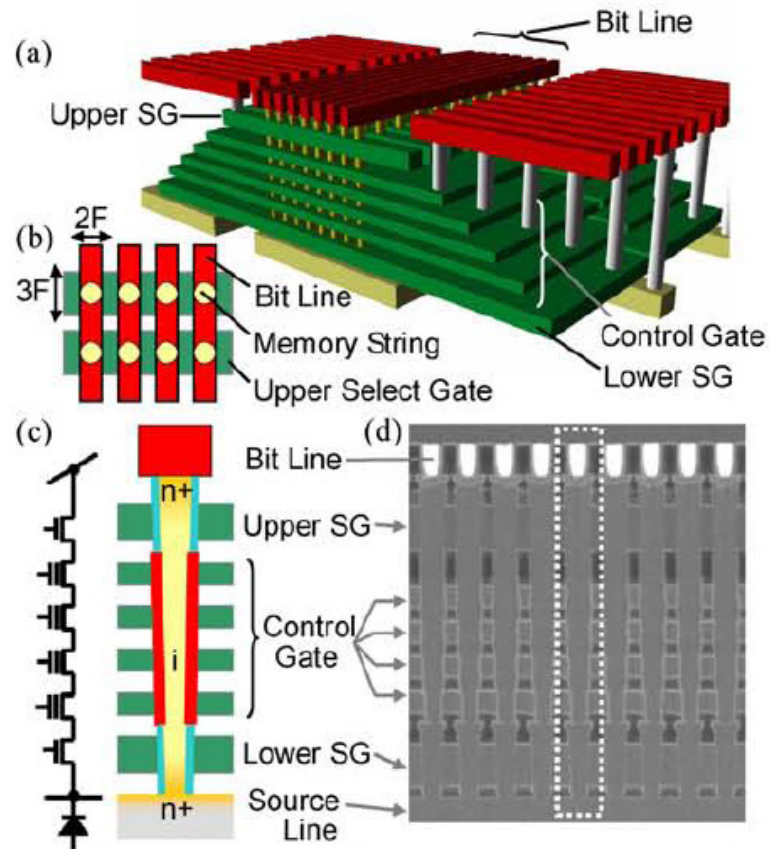


Figure 1: (a) Birds-eye view of BiCS Flash memory. (b) Top-down view of BiCS Flash memory array. (c) Enlarged view of the memory string. (d) Cross sectional SEM image of BiCS Flash memory array.

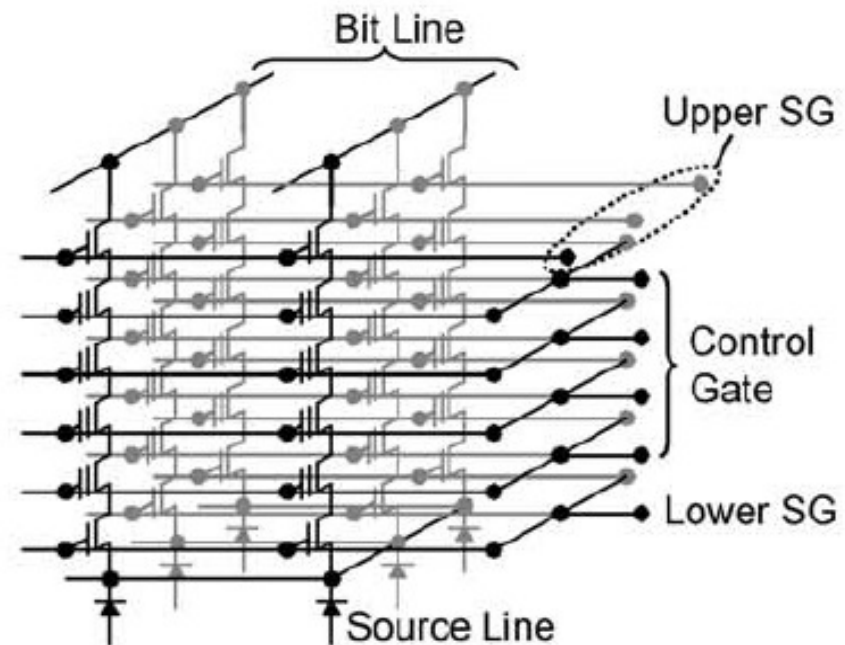
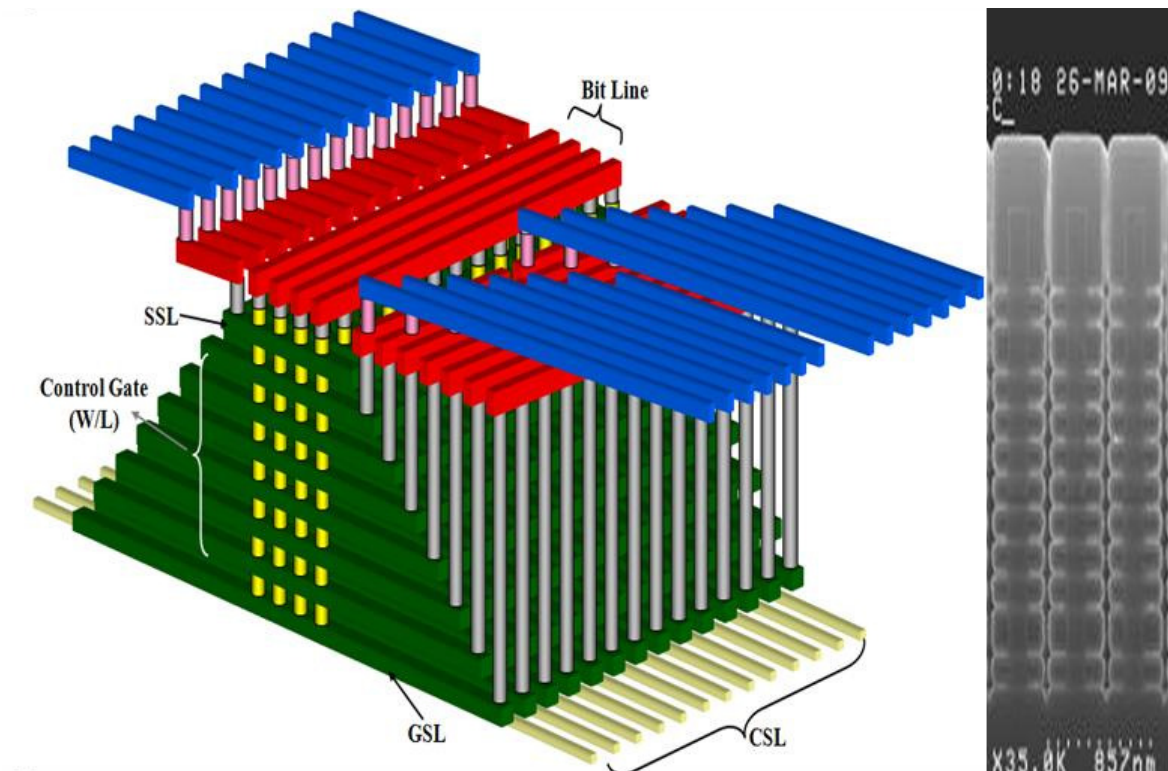


Figure 2: Equivalent circuit of BiCS Flash memory.

3D Flash memory

- TCAT(Terabit Cell Array Transistor) technology

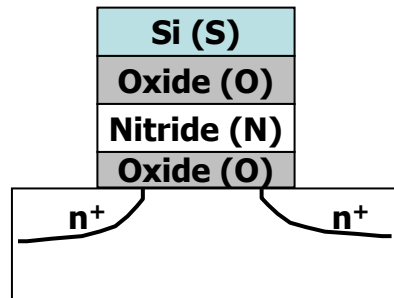
2009 VLSI, Samsung



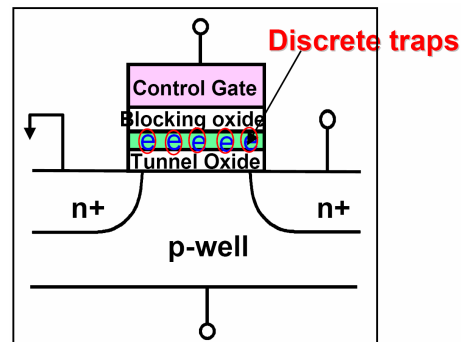
What is CTF ?

- CTF (Charge Trap Flash)
 - SONOS type Flash
 - electron is trapped in nitride trap layer

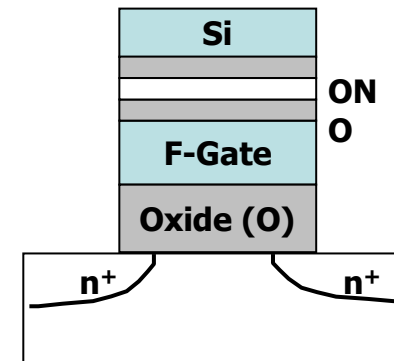
SONOS



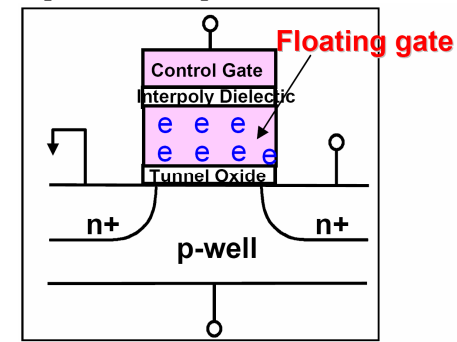
*P.C.Y. Chen, TED, V. 24,
pp.584-586, 1977*



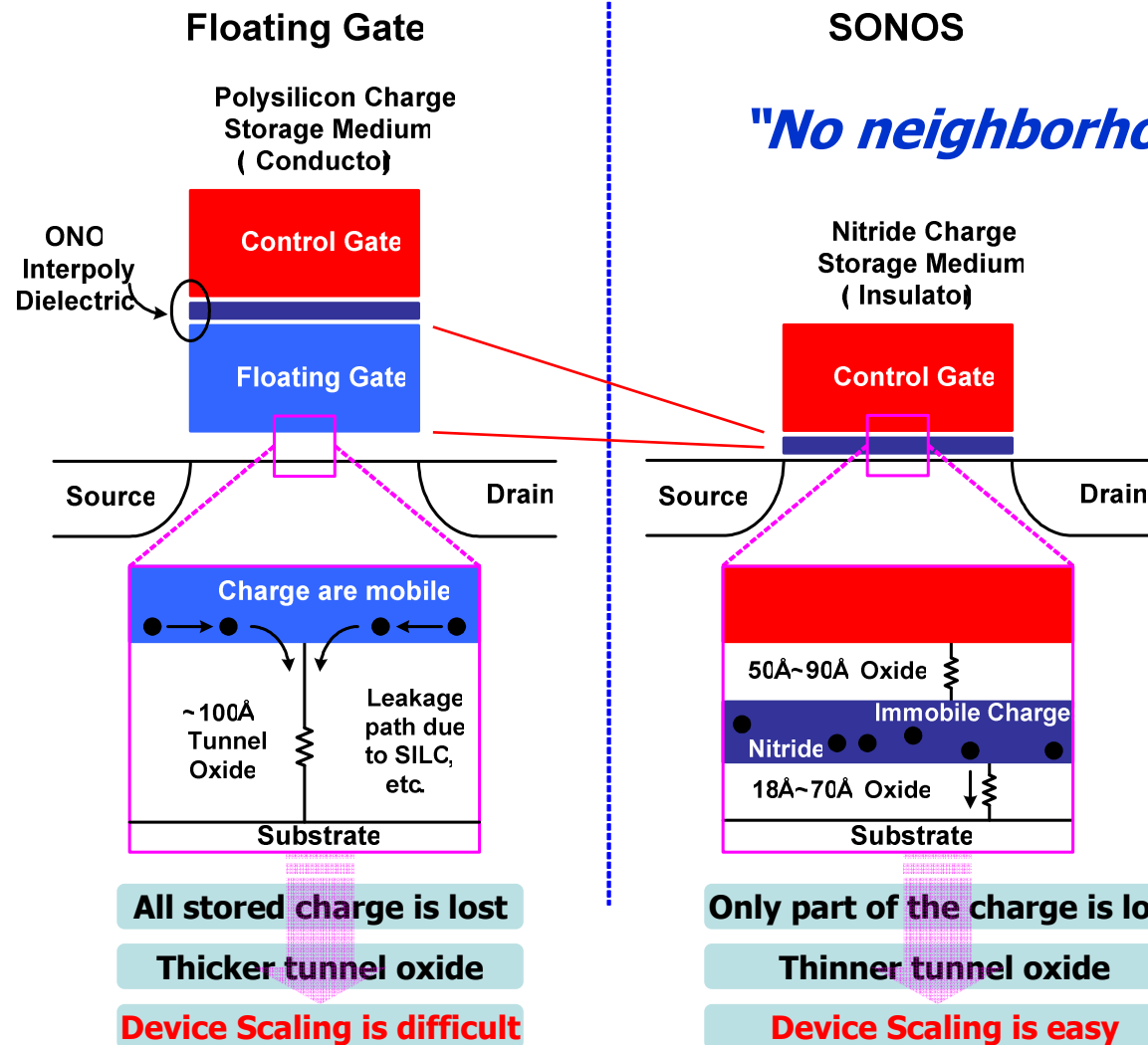
Floating Gate



*F.Masuoka
(Toshiba)*



Floating gate vs. SONOS



Thanks for your listening !

